Analisis Satelit, ERA5, dan AWS IoT (Jerukagung)

In [1]:
```python
# Set 1: Inisialisasi Import Semua Library & Dependensi
import os
import sys
import glob
import warnings
import numpy as np
import pandas as pd
import scipy.stats as stats
from scipy.stats import wilcoxon, mannwhitneyu
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import r2_score
from IPython.display import display, Markdown, HTML

# Set Directory Base & Hasil Analisis
base_dir = os.path.dirname(os.path.abspath(__file__)) if '__file__' in globals()
out_dir = os.path.join(base_dir, 'Hasil_Analisis')
os.makedirs(out_dir, exist_ok=True)

# Nonaktifkan warning berlebihan & atur tema grafik
warnings.filterwarnings('ignore')
plt.rcParams['figure.dpi'] = 600
sns.set_theme(style='whitegrid')
```

1. Inspeksi Data

In [2]:
```python
# Definisi Path Dataset
base_dir = r"d:/Github/Projek_Rainfall/Google_Earth_Engine"

file_paths = {
    'GSMaP': os.path.join(base_dir, 'Data_Satelit', 'Rainfall_GSMaP_TimeSeries_U
    'IMERG': os.path.join(base_dir, 'Data_Satelit', 'Rainfall_IMERG_TimeSeries_U
    'ERA5_Reanalisis': os.path.join(base_dir, 'Data_Satelit', 'ERA5_Hourly_All_R
    'AWS_Lokal': os.path.join(base_dir, 'Data_Satelit', 'id-05_clear_data_hourly
}

# Fungsi deteksi otomatis untuk semua variabel
def detect_columns(df):
    cols = [c.lower() for c in df.columns]

    # Detect Time
    dt_col, unix_col = None, None
    for c in df.columns:
        cl = c.lower()
        if 'unix' in cl and unix_col is None:
            unix_col = c
        elif ('date' in cl or 'time' in cl) and ('unix' not in cl) and dt_col is
            dt_col = c

    # Detect Variables
    var_keywords = {
```

```
        'rain': ['rainfall', 'precipitation', 'precip', 'rain', 'rainrate', 'tot
        'temperature': ['temperature', 'temp'],
        'humidity': ['humidity', 'rh', 'relative_humidity'],
        'pressure': ['pressure', 'pres'],
        'dewpoint': ['dew_point', 'dewpoint', 'dew']
    }

    var_cols = {}
    for var, keywords in var_keywords.items():
        found_cols = []
        for c in df.columns:
            cl = c.lower()
            if any(k in cl for k in keywords):
                # Prioritaskan Gauge-Calibrated jika ada (untuk hujan)
                if 'gc' in cl:
                    found_cols.insert(0, c)
                else:
                    found_cols.append(c)
        if found_cols:
            var_cols[var] = found_cols[0]

    return dt_col, unix_col, var_cols

datasets_raw = {}
summary_data = []


for name, path_file in file_paths.items():
    if not os.path.exists(path_file):
        print(f"File not found: {path_file}")
        continue

    if path_file.endswith('.csv'):
        df = pd.read_csv(path_file)
    elif path_file.endswith('.parquet'):
        df = pd.read_parquet(path_file)

    dt_col, unix_col, var_cols = detect_columns(df)

    # Kalkulasi resolusi
    resolution = "Unknown"
    start_date, end_date = "Unknown", "Unknown"
    if unix_col and not df.empty:
        df = df.sort_values(unix_col)
        diffs = df[unix_col].diff().dropna()
        mode_diff = diffs.mode().iloc[0] if not diffs.empty else 0
        if mode_diff == 3600:
            resolution = "Hourly"
        elif mode_diff == 1800:
            resolution = "30-Minute"
        elif mode_diff == 86400:
            resolution = "Daily"
        else:
            resolution = f"{mode_diff} seconds"

    if dt_col and not df.empty:
        start_date = str(df[dt_col].iloc[0])
        end_date = str(df[dt_col].iloc[-1])

    summary_data.append({
```

```
        'Dataset': name,
        'Records': len(df),
        'Datetime Field': dt_col,
        'Unix Field': unix_col,
        'Variables Found': ", ".join(var_cols.keys()),
        'Resolution': resolution,
        'Start': start_date,
        'End': end_date
    })

    datasets_raw[name] = df

# Generate Markdown Table
md_table = "| Dataset | Records | Datetime Field | Unix Field | Variables Found
md_table += "|---|---|---|---|---|---|---|---|\n"
for row in summary_data:
    md_table += f"| {row['Dataset']} | {row['Records']:,} | `{row['Datetime Fiel

display(Markdown("### Data Inspection Summary\n" + md_table))
```

Data Inspection Summary

Dataset	Records	Datetime Field	Unix Field	Variables Found	Resolution	Start	
GSMaP	250,283	`datetime_utc`	`unixtime`	rain	3600000.0 seconds	1998-01-01 00:00:00	2 0 1
IMERG	500,548	`datetime_utc`	`unixtime`	rain	1800000.0 seconds	1998-01-01 00:00:00	2 0 0
ERA5_Reanalisis	249,768	`datetime_utc`	`unixtime`	rain, temperature, humidity, pressure, dewpoint	3600000.0 seconds	1998-01-01 00:00:00	2 0 2
AWS_Lokal	13,512	`datetime_utc`	`unixtime`	rain, temperature, humidity, pressure, dewpoint	Hourly	2025-01-01 00:00:00	2 0 2

2. Standarisasi Waktu

In [3]:
```
processed_dfs = {}
validation_logs = []

for name, df in datasets_raw.items():
    df_proc = df.copy()
    dt_col, unix_col, var_cols = detect_columns(df_proc)

    if not unix_col or not var_cols:
        validation_logs.append(f"- **{name}**: Failed. Missing Unix or any known
        continue
```

```
    # Standardize time
    df_proc = df_proc.dropna(subset=[unix_col] + list(var_cols.values()), how='a
    df_proc[unix_col] = df_proc[unix_col].astype('float64')

    if df_proc[unix_col].max() > 1e11:
        df_proc[unix_col] = df_proc[unix_col] / 1000.0

    # Cast unixtime to datetime strictly as UTC
    df_proc['standard_time'] = pd.to_datetime(df_proc[unix_col], unit='s', utc=T
    df_proc = df_proc.set_index('standard_time')

    # Extract only the detected variables
    df_vars = df_proc[list(var_cols.values())].copy()

    # Rename columns to standardized format (e.g., rain_GSMaP, temp_ERA5_ML)
    rename_mapping = {col_name: f"{var_type}_{name}" for var_type, col_name in v
    df_vars = df_vars.rename(columns=rename_mapping)

    validation_logs.append(f"- **{name}**: Kept variables: {', '.join(var_cols.k

    # METEOROLOGICAL LOGIC: Align all datasets to 1-Hour intervals (1h)
    # Rain is accumulated (sum), other variables are averaged (mean)
    resample_dict = {}
    for col in df_vars.columns:
        if col.startswith('rain_'):
            resample_dict[col] = 'sum'
        else:
            resample_dict[col] = 'mean'

    if not df_vars.empty:
        df_vars = df_vars.resample('1h').agg(resample_dict)

    processed_dfs[name] = df_vars.dropna(how='all')

display(Markdown("### Temporal Standardization Logs\n" + "\n".join(validation_lo
```

Temporal Standardization Logs

- **GSMaP**: Kept variables: rain. No temporal resampling applied (exact match only).
- **IMERG**: Kept variables: rain. No temporal resampling applied (exact match only).
- **ERA5_Reanalisis**: Kept variables: rain, temperature, humidity, pressure, dewpoint. No temporal resampling applied (exact match only).
- **AWS_Lokal**: Kept variables: rain, temperature, humidity, pressure, dewpoint. No temporal resampling applied (exact match only).

3. Penggabungan Data

In [4]:
```
# Merge all datasets on standard_time (Index)
df_merged = None

for name, df in processed_dfs.items():
    if df_merged is None:
        df_merged = df
    else:
        df_merged = df_merged.merge(df, left_index=True, right_index=True, how='
```

```
# Drop rows where ALL values are NA
df_merged = df_merged.dropna(how='all')

# Restrict analysis timeline to 2025 onwards
df_merged = df_merged[(df_merged.index >= '2025-01-01 00:00:00') & (df_merged.in

overlap_start = df_merged.index.min()
overlap_end = df_merged.index.max()
overlap_duration = overlap_end - overlap_start

display(Markdown(f"### Overlap Alignment Summary\n"
                 f"- **Overlap Duration**: {overlap_duration}\n"
                 f"- **Overlapping Record Count**: {len(df_merged):,}\n"
                 f"- **Start Overlap**: {overlap_start}\n"
                 f"- **End Overlap**: {overlap_end}"))

df_merged.head()
```

Overlap Alignment Summary

- **Overlap Duration**: 515 days 23:00:00
- **Overlapping Record Count**: 12,384
- **Start Overlap**: 2025-01-01 00:00:00+00:00
- **End Overlap**: 2026-05-31 23:00:00+00:00

Out[4]:

standard_time	rain_GSMaP	rain_IMERG	rain_ERA5_Reanalisis	temperature_ERA5_Reana
2025-01-01 00:00:00+00:00	0.0	0.0	0.398636	26.217
2025-01-01 01:00:00+00:00	0.0	0.0	0.208855	27.214
2025-01-01 02:00:00+00:00	0.0	0.0	0.475407	29.295
2025-01-01 03:00:00+00:00	0.0	0.0	0.238419	29.890
2025-01-01 04:00:00+00:00	0.0	0.0	0.305653	30.245

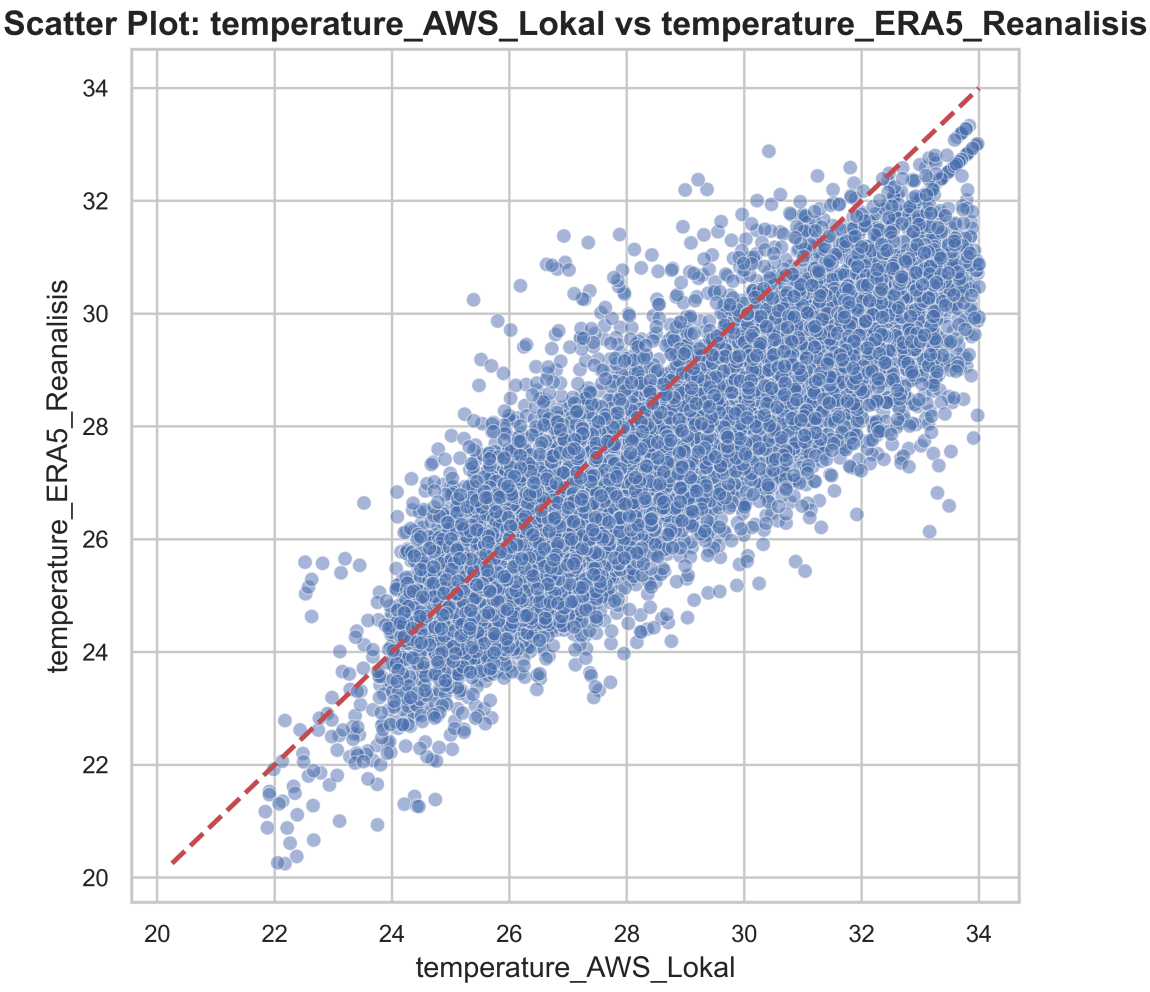
4. Analisis Korelasi

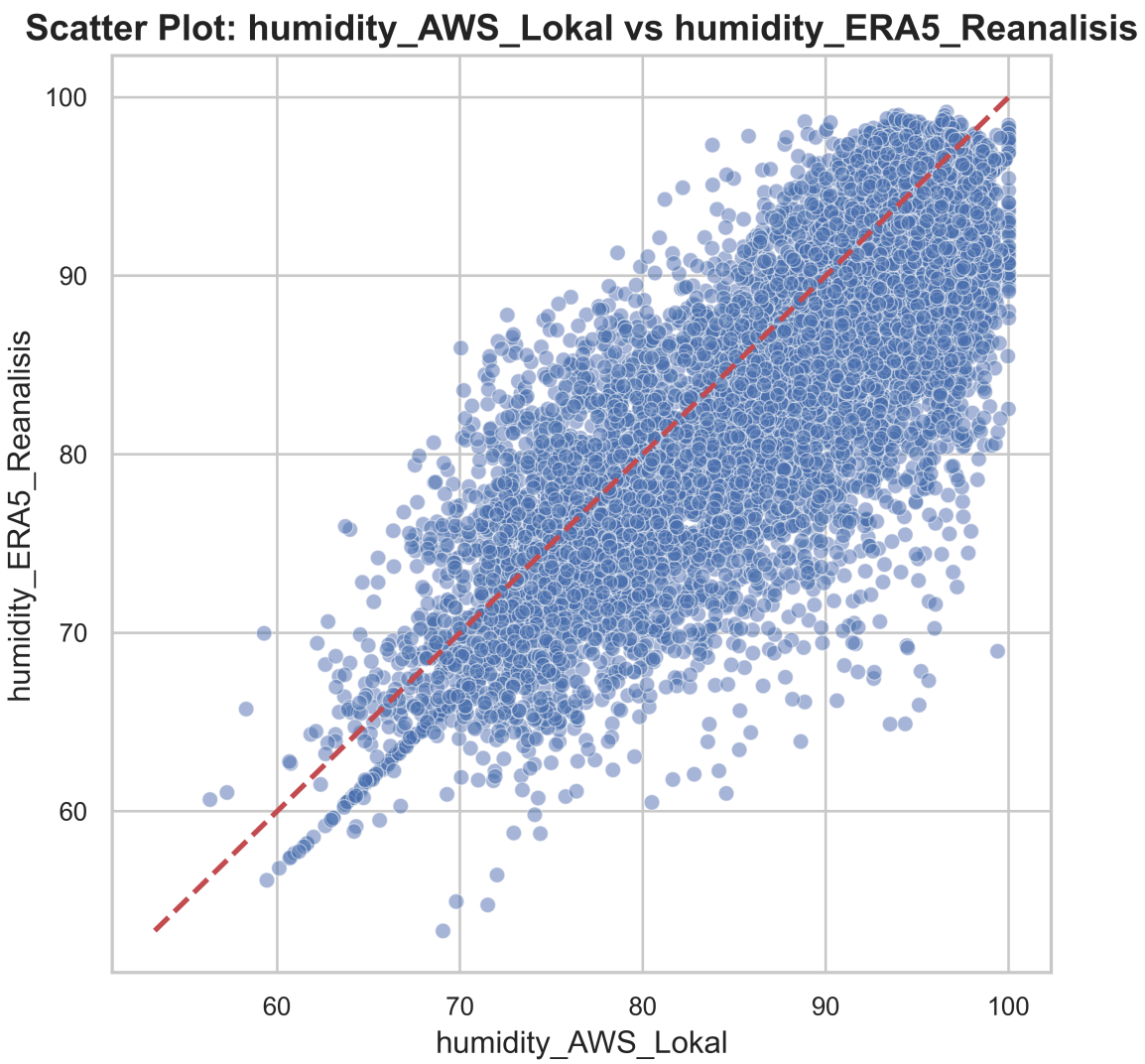
In [5]:
```
# Sel 9.1: Inisialisasi Variabel Analisis
var_types = ['temperature', 'humidity', 'pressure', 'dewpoint', 'rain']
print(" Variabel analisis diinisialisasi:", var_types)
```

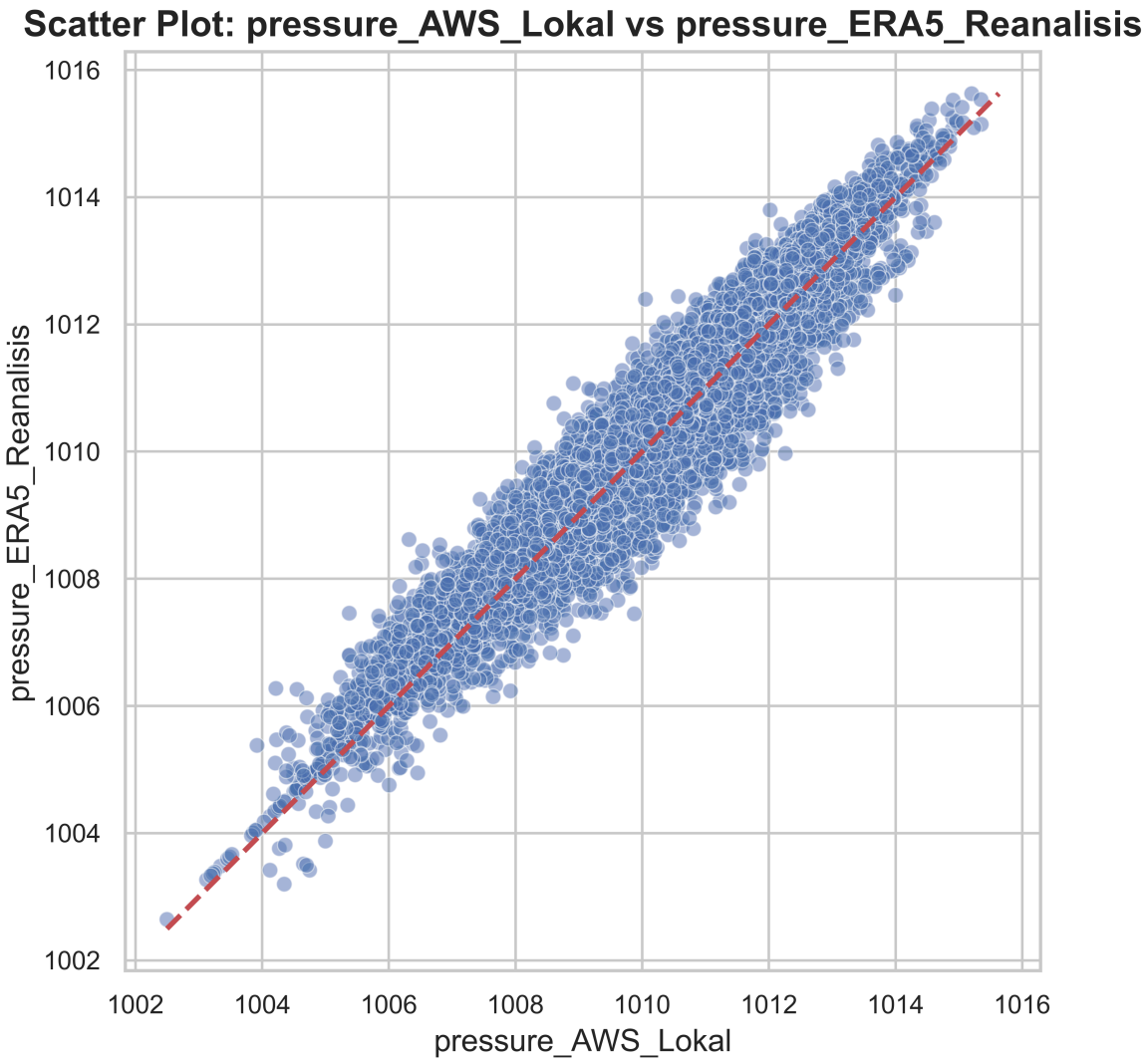
```
 Variabel analisis diinisialisasi: ['temperature', 'humidity', 'pressure', 'dewpo
int', 'rain']
```

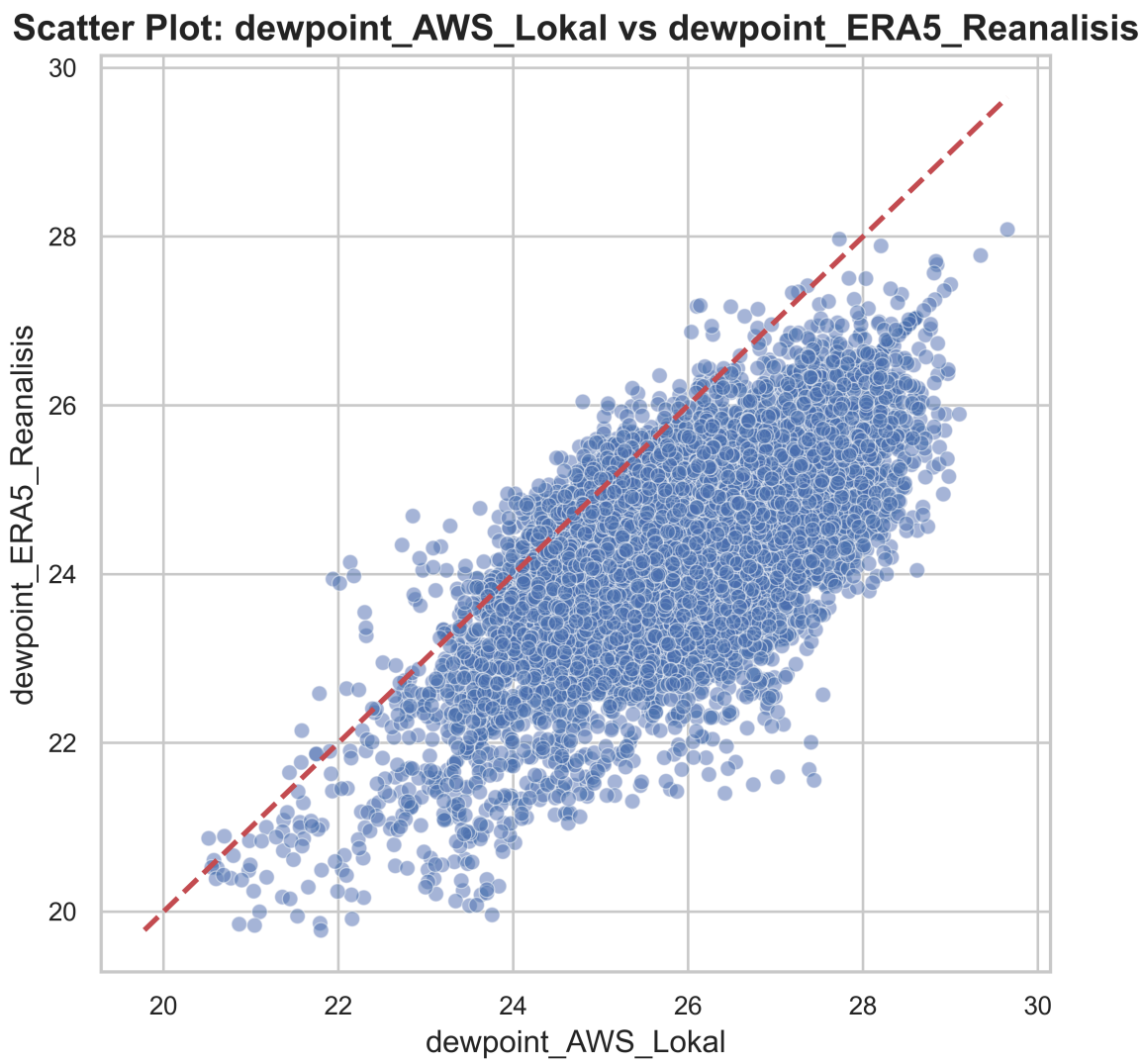
In [6]:
```
# Sel 9.5: Scatter Plots vs Sensor IoT
for var in var_types:
    var_cols = [c for c in df_merged.columns if c.startswith(f'{var}_')]
    if len(var_cols) < 2: continue
```

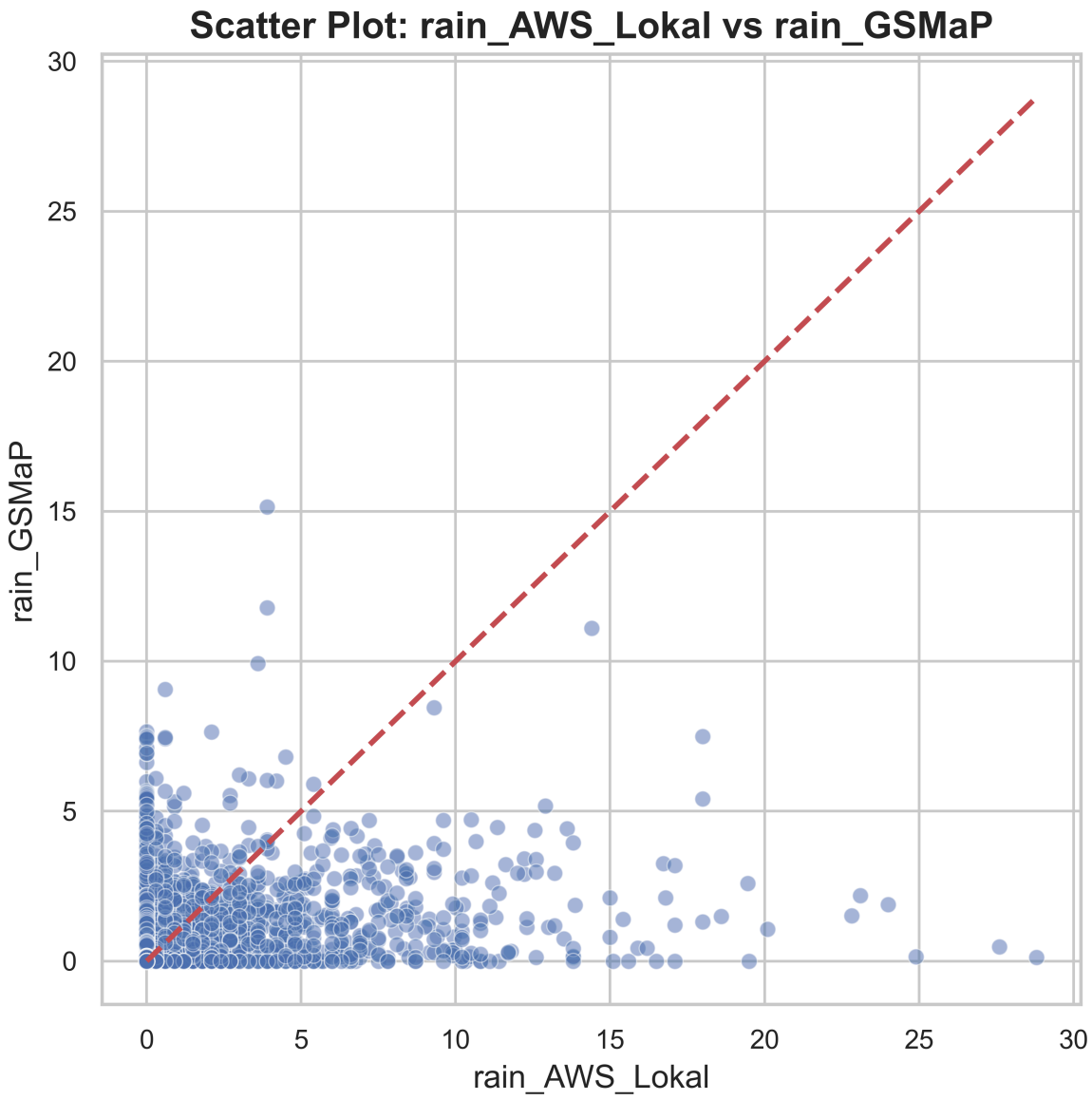
```
    iot_col = None
    for cand in [f'{var}_AWS_Lokal', f'{var}_IoT_curah_hujan', f'{var}_AWS', f'{
        if cand in df_merged.columns:
            iot_col = cand
            break

    if iot_col:
        compare_cols = [c for c in var_cols if c != iot_col]
        for col in compare_cols:
            plt.figure(figsize=(6, 6))
            sns.scatterplot(x=df_merged[iot_col], y=df_merged[col], alpha=0.5)

            # Perfect fit line
            min_val = min(df_merged[iot_col].min(), df_merged[col].min())
            max_val = max(df_merged[iot_col].max(), df_merged[col].max())
            plt.plot([min_val, max_val], [min_val, max_val], 'r--', linewidth=2.

            plt.title(f'Scatter Plot: {iot_col} vs {col}', fontsize=14, fontweig
            plt.xlabel(iot_col, fontsize=12)
            plt.ylabel(col, fontsize=12)
            plt.xticks(fontsize=10)
            plt.yticks(fontsize=10)
            plt.tight_layout()
            plt.savefig(os.path.join(out_dir, 'scatter_vs_iot_{var}_{col}.png'),
            plt.show()
```

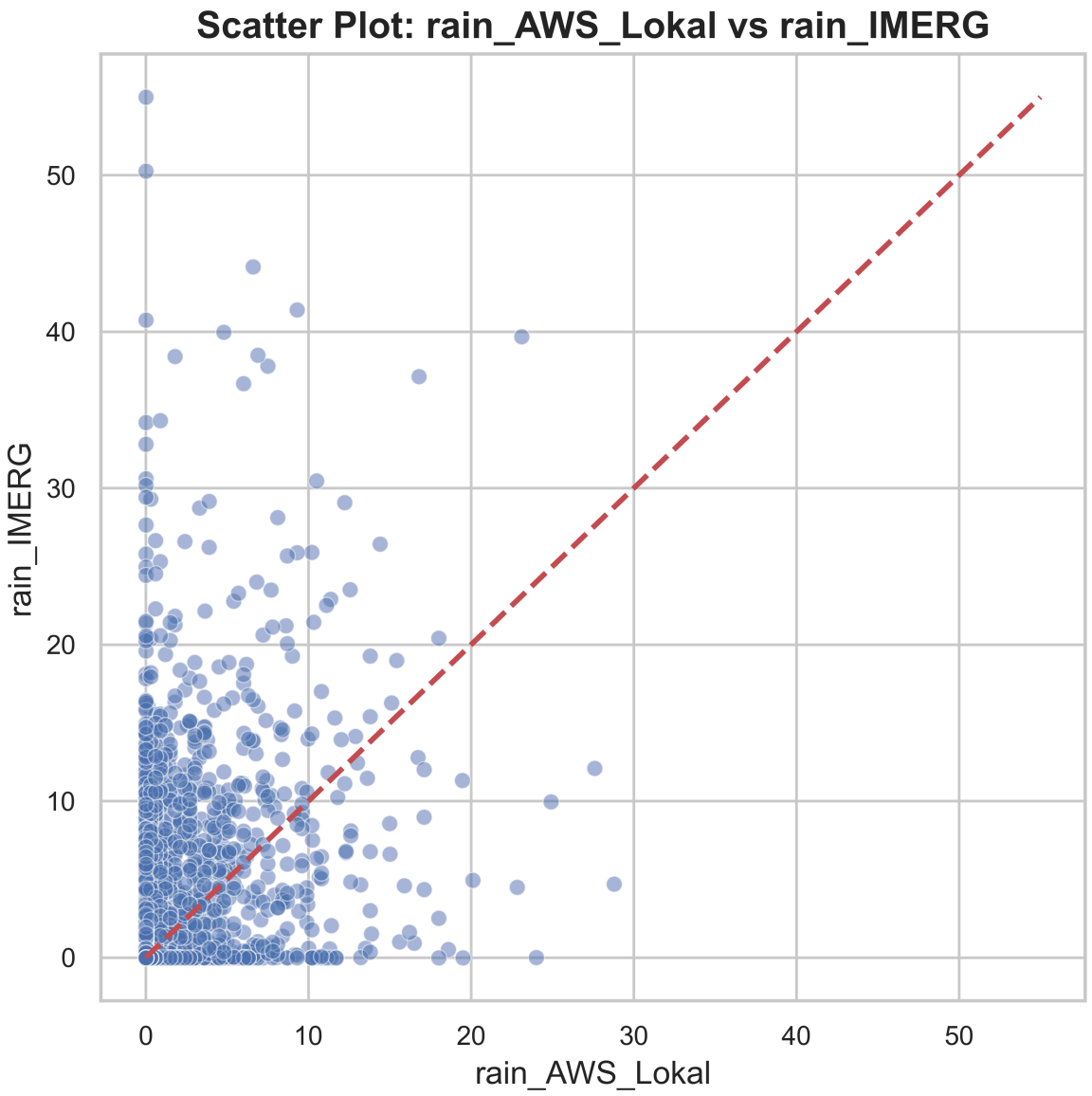


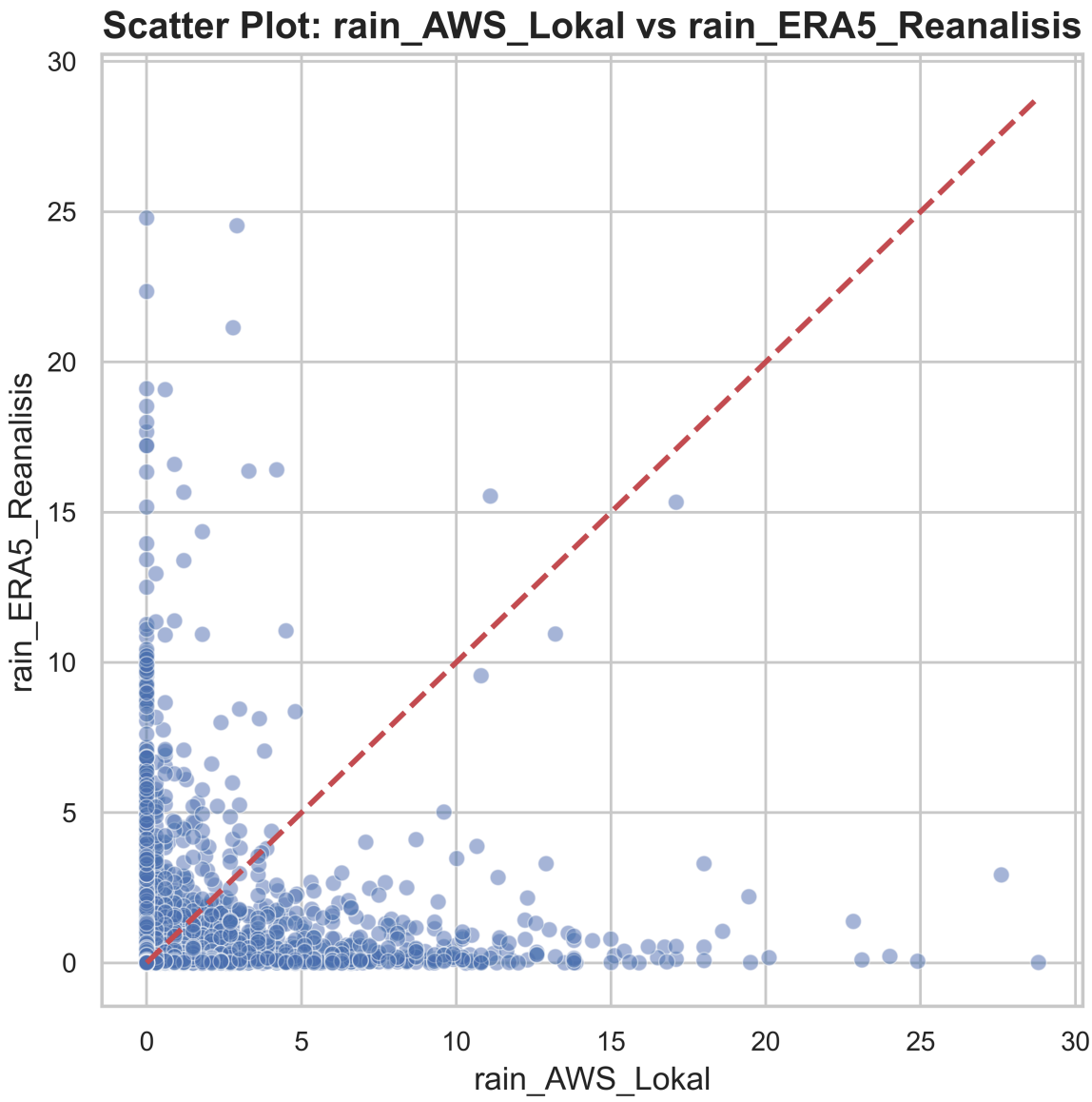

Scatter Plot: humidity_AWS_Lokal vs humidity_ERA5_Reanalisis
humidity_ERA5_Reanalisis
100
90
80
70
60
60
70
80
90
100
humidity_AWS_Lokal


Scatter Plot: pressure_AWS_Lokal vs pressure_ERA5_Reanalisis
pressure_ERA5_Reanalisis
1016
1014
1012
1010
1008
1006
1004
1002
1002
1004
1006
1008
1010
1012
1014
1016
pressure_AWS_Lokal

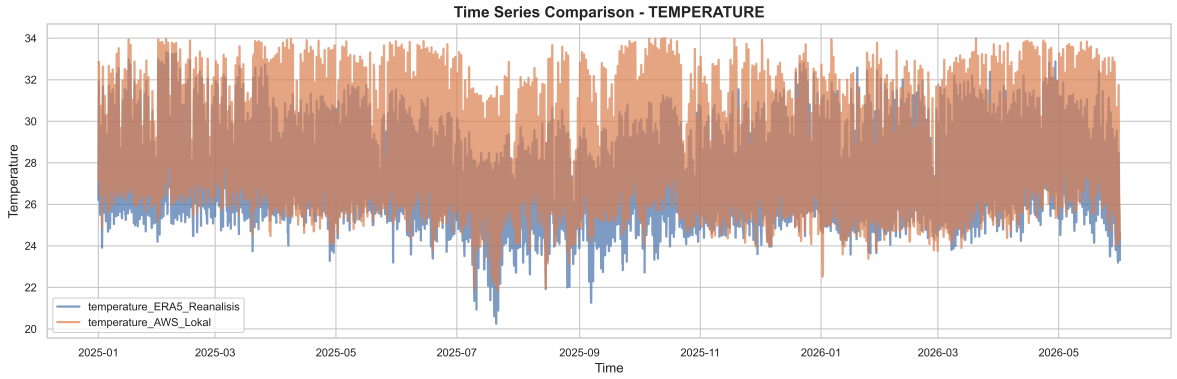

Scatter Plot: dewpoint_AWS_Lokal vs dewpoint_ERA5_Reanalisis
dewpoint_ERA5_Reanalisis
dewpoint_AWS_Lokal
30
28
26
24
22
20
20
22
24
26
28
30

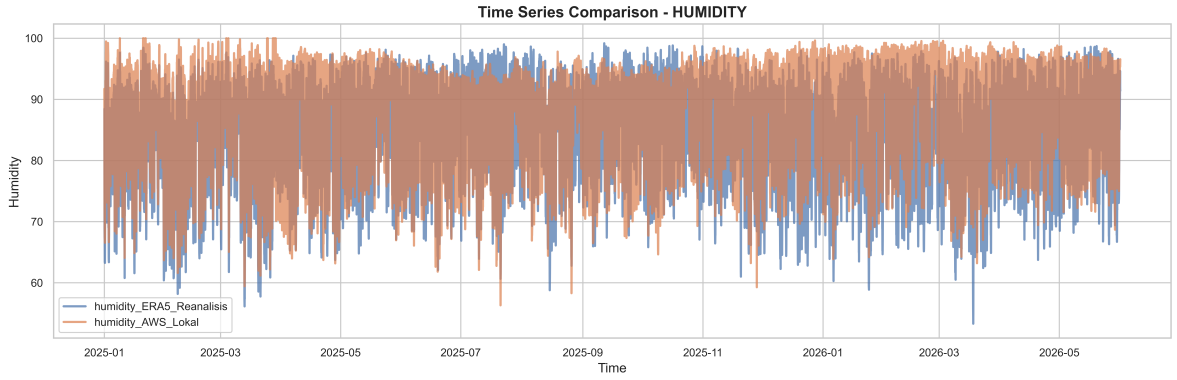

Scatter Plot: rain_AWS_Lokal vs rain_GSMaP
rain_GSMaP
rain_AWS_Lokal
0
5
10
15
20
25
30

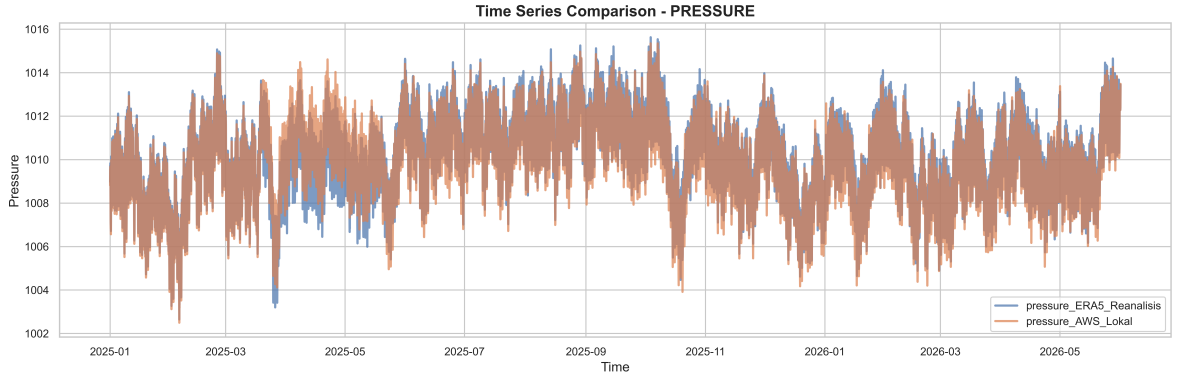

Scatter Plot: rain_AWS_Lokal vs rain_IMERG
rain_IMERG
rain_AWS_Lokal
0
10
20
30
40
50

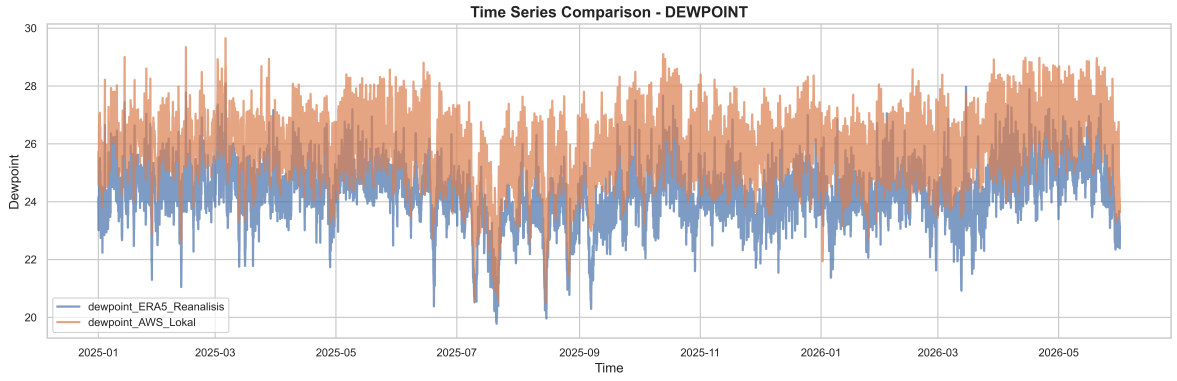


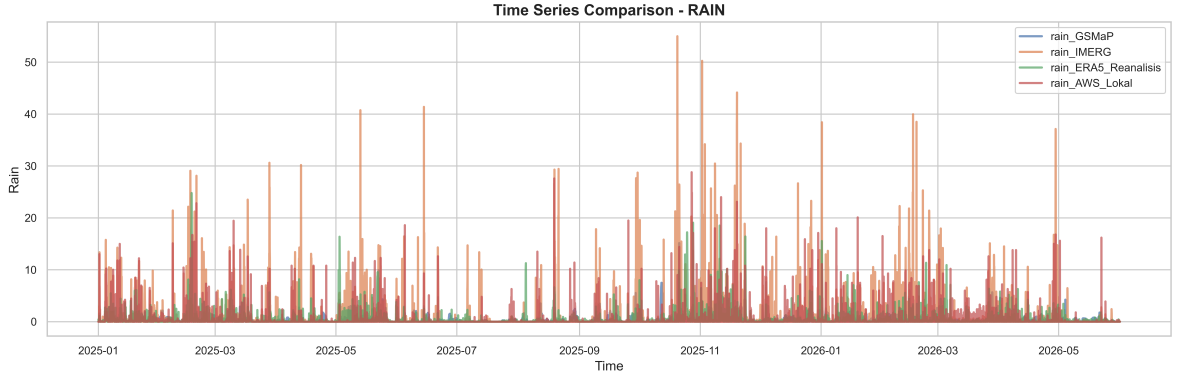
In [7]:
```python
# Sel 9.6: Perbandingan Time Series
for var in var_types:
    var_cols = [c for c in df_merged.columns if c.startswith(f'{var}_')]
    if len(var_cols) < 2: continue

    plt.figure(figsize=(15, 5))
    for col in var_cols:
        plt.plot(df_merged.index, df_merged[col], label=col, alpha=0.7, linewidt
    plt.title(f'Time Series Comparison - {var.upper()}', fontsize=14, fontweight
    plt.xlabel('Time', fontsize=12)
    plt.ylabel(var.capitalize(), fontsize=12)
    plt.xticks(fontsize=10)
    plt.yticks(fontsize=10)
    plt.legend(fontsize=10)
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'timeseries_{var}.png'), dpi=600, bbox_inc
    plt.show()
```


Time Series Comparison - TEMPERATURE
Temperature
Time
temperature_ERA5_Reanalisis
temperature_AWS_Lokal


Time Series Comparison - HUMIDITY
Humidity
Time
humidity_ERA5_Reanalisis
humidity_AWS_Lokal

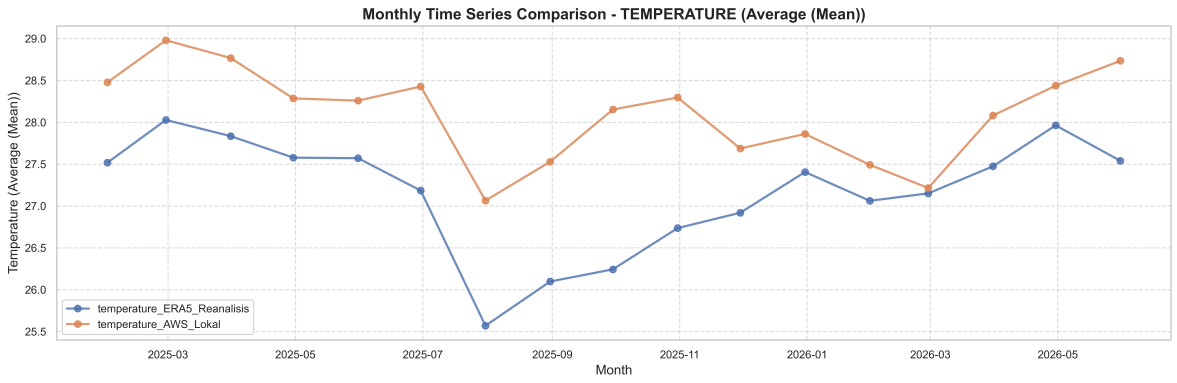

Time Series Comparison - PRESSURE
Pressure
Time
pressure_ERA5_Reanalisis
pressure_AWS_Lokal

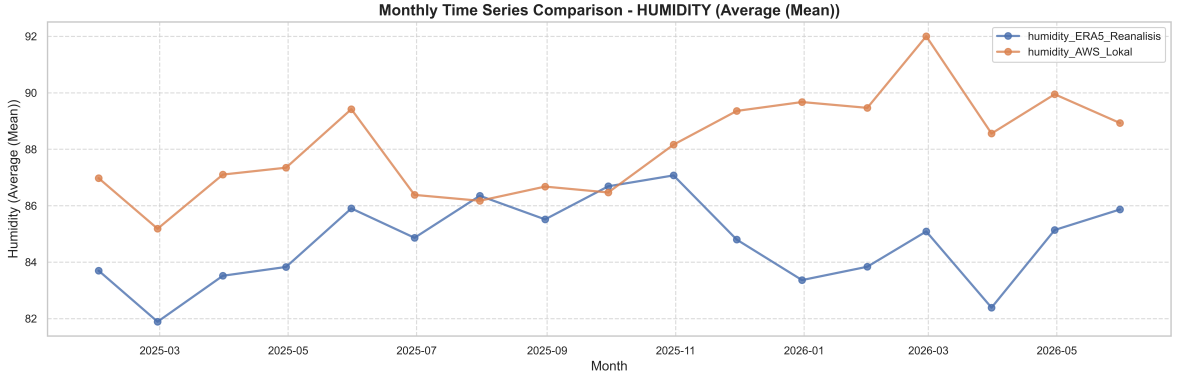

Time Series Comparison - DEWPOINT
Dewpoint
Time
dewpoint_ERA5_Reanalisis
dewpoint_AWS_Lokal

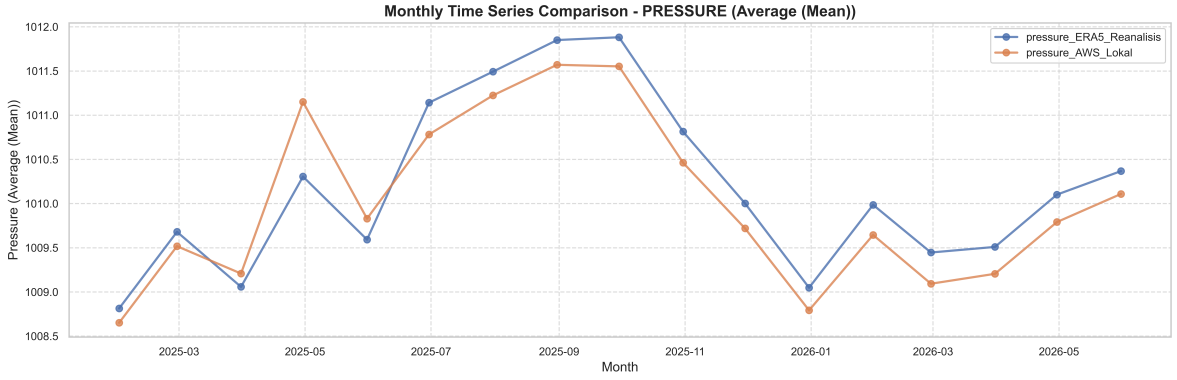

Time Series Comparison - RAIN
Rain
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

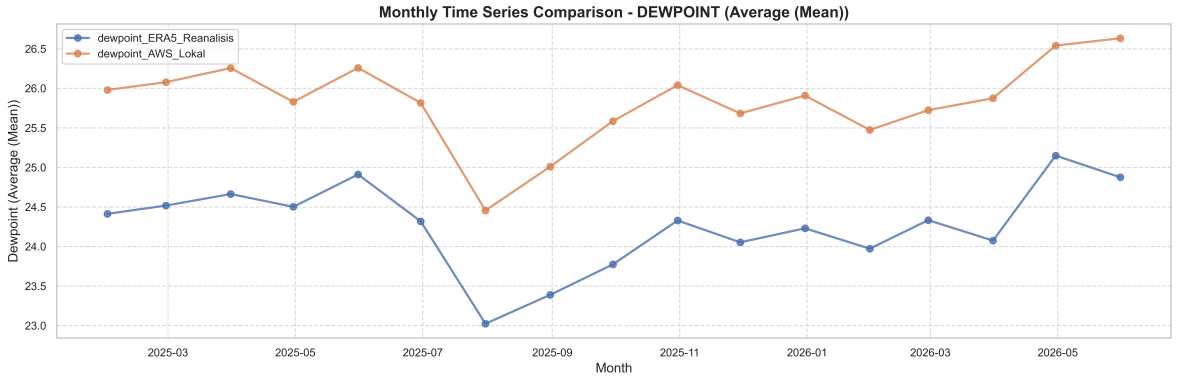
In [8]:

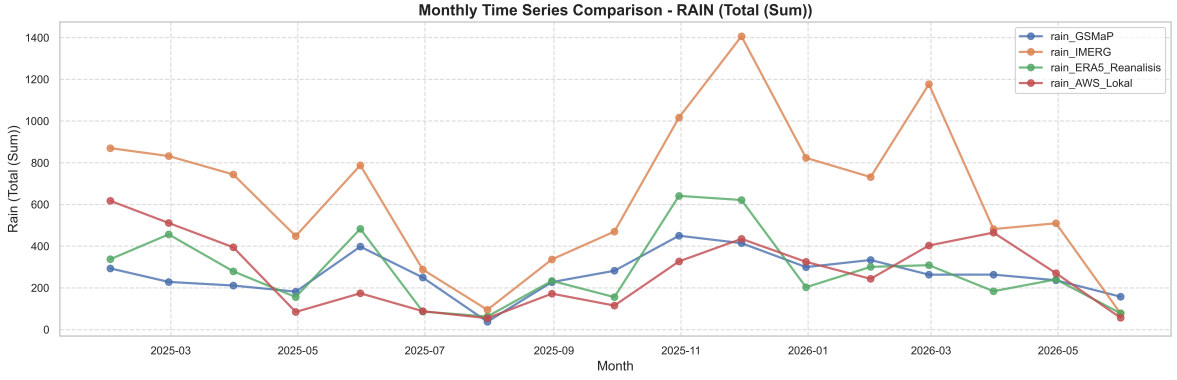
```
# Sel 9.7: Perbandingan Bulanan Time Series & Hourly Rainfall per Month
for var in var_types:
    var_cols = [c for c in df_merged.columns if c.startswith(f'{var}_')]
    if len(var_cols) < 2: continue

    plt.figure(figsize=(15, 5))
    if var == 'rain':
        df_monthly = df_merged[var_cols].resample('ME').sum()
        agg_label = 'Total (Sum)'
    else:
        df_monthly = df_merged[var_cols].resample('ME').mean()
        agg_label = 'Average (Mean)'

    for col in var_cols:
        plt.plot(df_monthly.index, df_monthly[col], marker='o', label=col, alpha
    plt.title(f'Monthly Time Series Comparison - {var.upper()} ({agg_label})', f
    plt.xlabel('Month', fontsize=12)
    plt.ylabel(f"{var.capitalize()} ({agg_label})", fontsize=12)
    plt.xticks(fontsize=10)
    plt.yticks(fontsize=10)
    plt.legend(fontsize=10)
    plt.grid(True, linestyle='--', alpha=0.6)
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'timeseries_monthly_{var}.png'), dpi=600,
    plt.show()

    # 7. Rain: Monthly Subplots (Hourly resolution, split by month)
    if var == 'rain':
        display(Markdown("#### Hourly Time Series per Month (Rainfall only)"))
        groups = df_merged[var_cols].groupby(df_merged.index.to_period('M'))

        for m, df_m in groups:
            if df_m.dropna(how='all').empty: continue

            plt.figure(figsize=(15, 4))
            for col in var_cols:
                valid_data = df_m[col].dropna()
                if not valid_data.empty:
                    plt.plot(valid_data.index, valid_data.values, label=col, alp

            plt.title(f'Hourly Rainfall - {m}', fontsize=14, fontweight='bold')
            plt.xlabel('Time', fontsize=12)
            plt.ylabel('Rainfall (mm)', fontsize=12)
            plt.xticks(fontsize=10)
            plt.yticks(fontsize=10)
            plt.legend(loc='center left', bbox_to_anchor=(1, 0.5), fontsize=10)
            plt.grid(True, linestyle='--', alpha=0.5)
            plt.tight_layout()
            plt.savefig(os.path.join(out_dir, 'hourly_rain_{m}.png'), dpi=600, b
            plt.show()
```



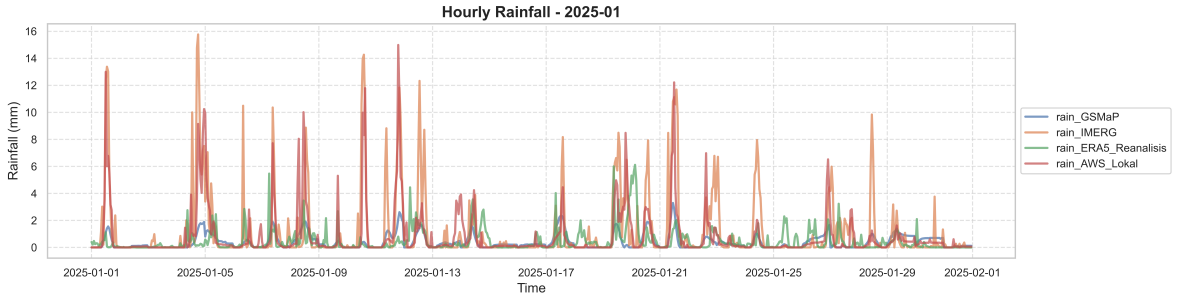


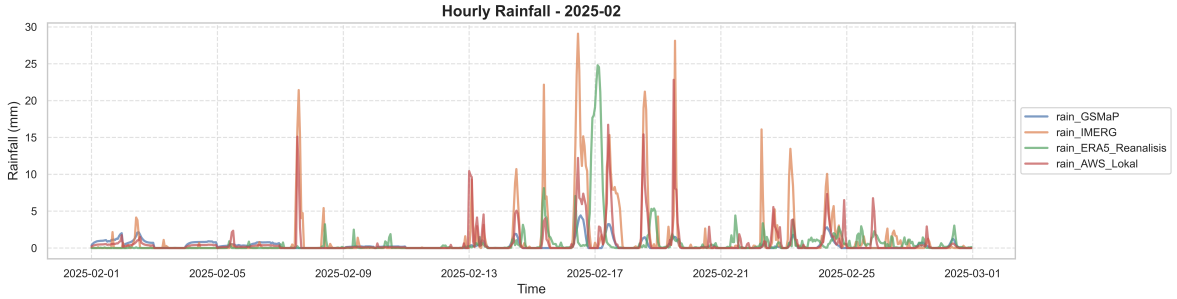


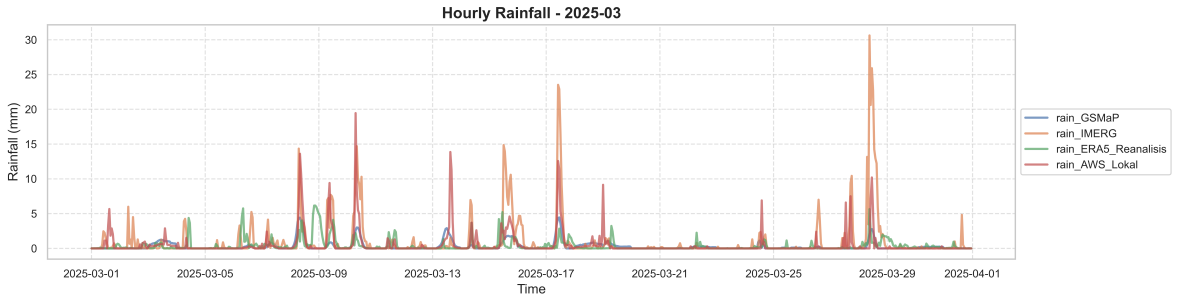


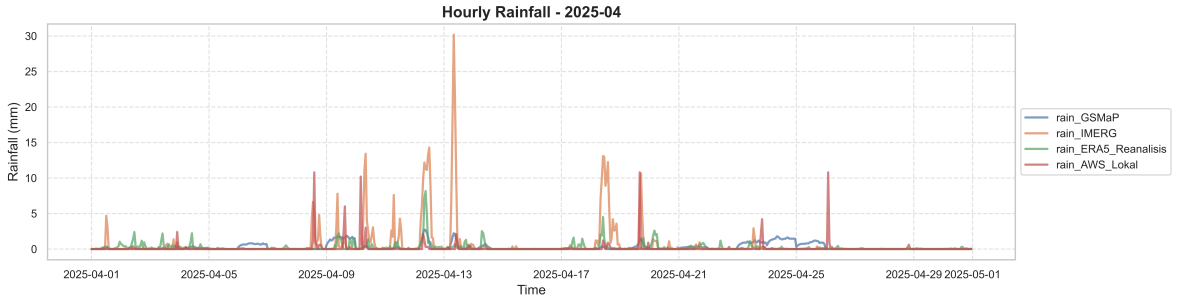


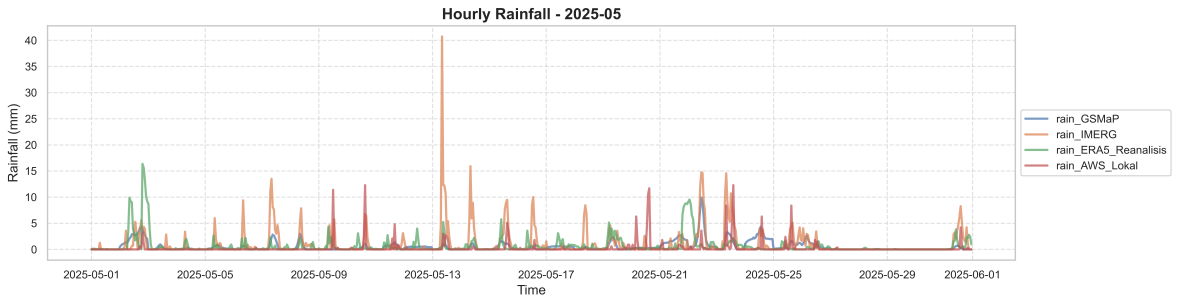
Hourly Time Series per Month (Rainfall only)

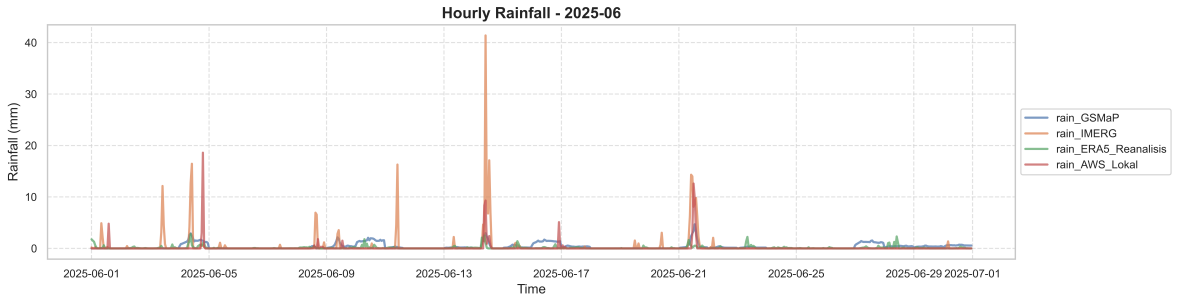

Hourly Rainfall - 2025-01
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

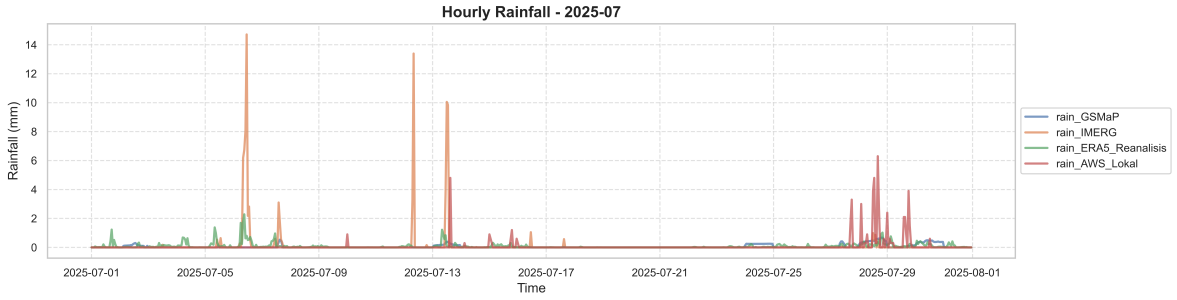

Hourly Rainfall - 2025-02
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

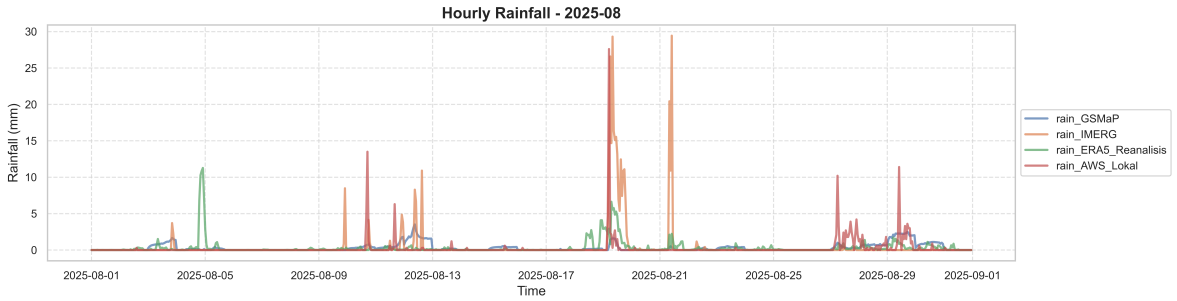

Hourly Rainfall - 2025-03
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

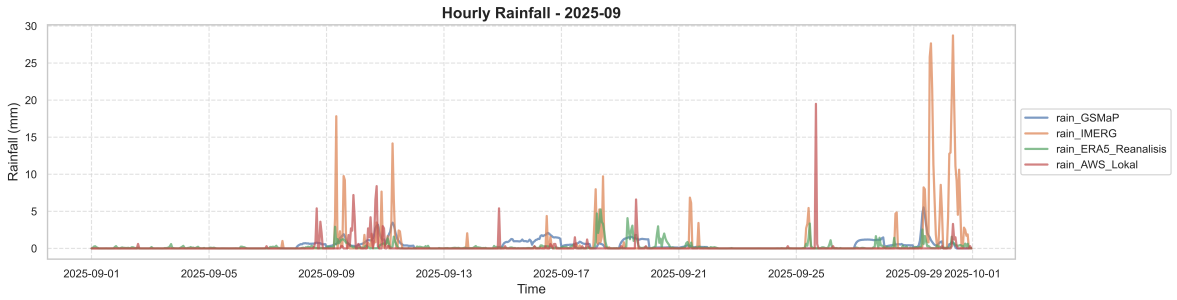

Hourly Rainfall - 2025-04
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

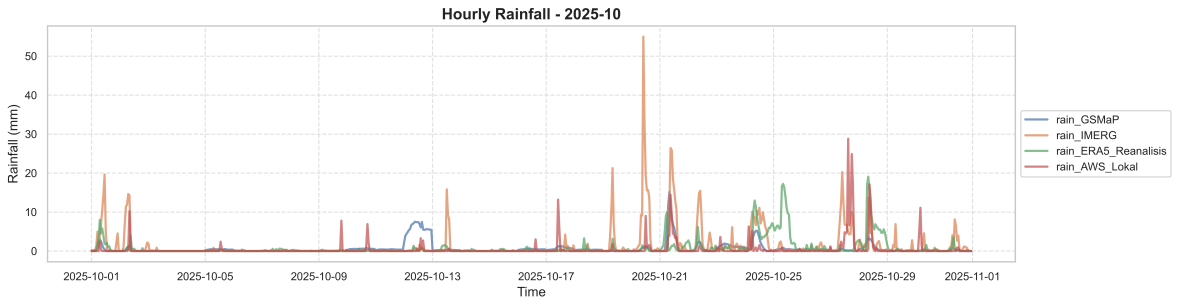

Hourly Rainfall - 2025-05
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

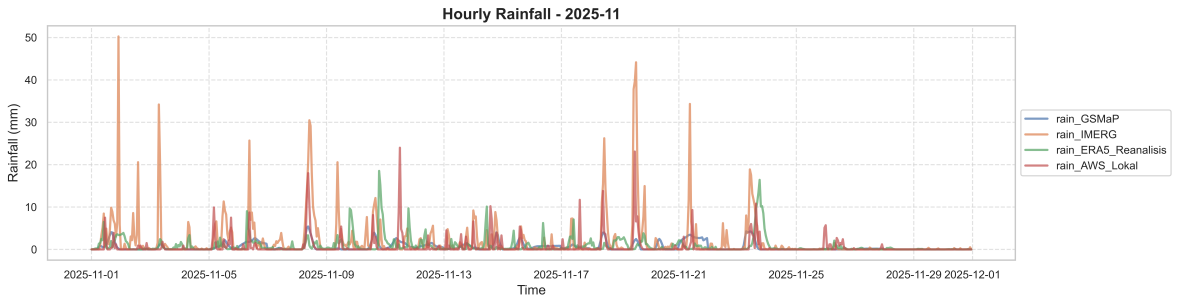

Hourly Rainfall - 2025-06
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

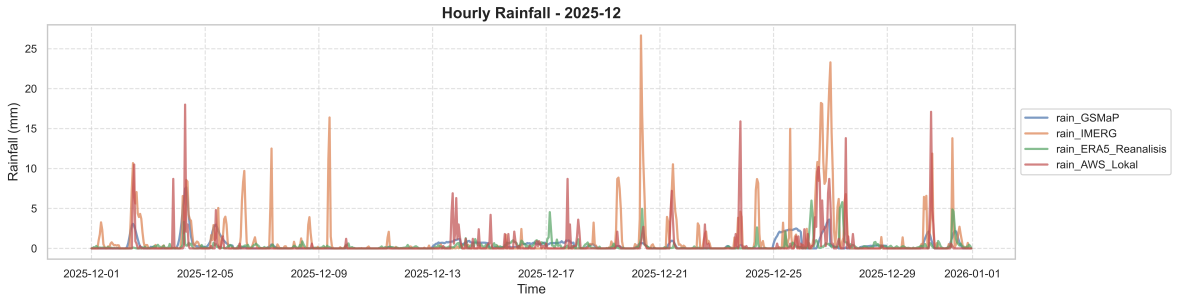

Hourly Rainfall - 2025-07
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

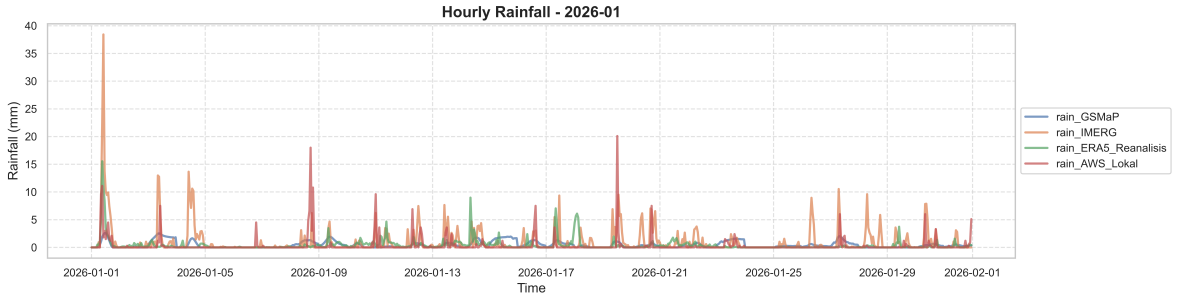

Hourly Rainfall - 2025-08
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

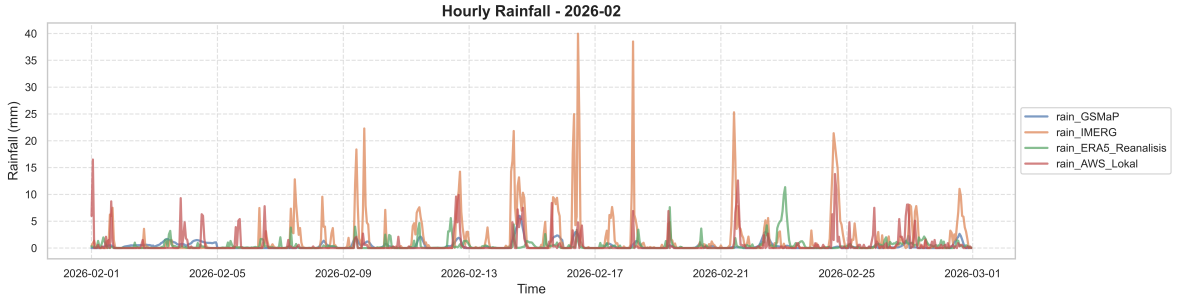

Hourly Rainfall - 2025-09
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

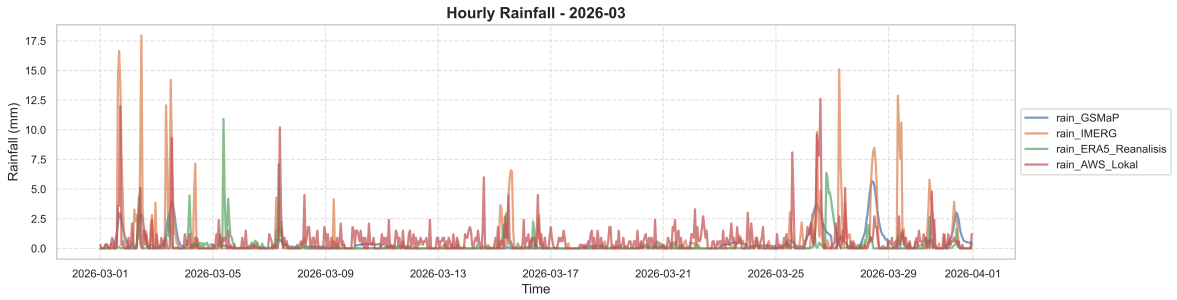

Hourly Rainfall - 2025-10
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

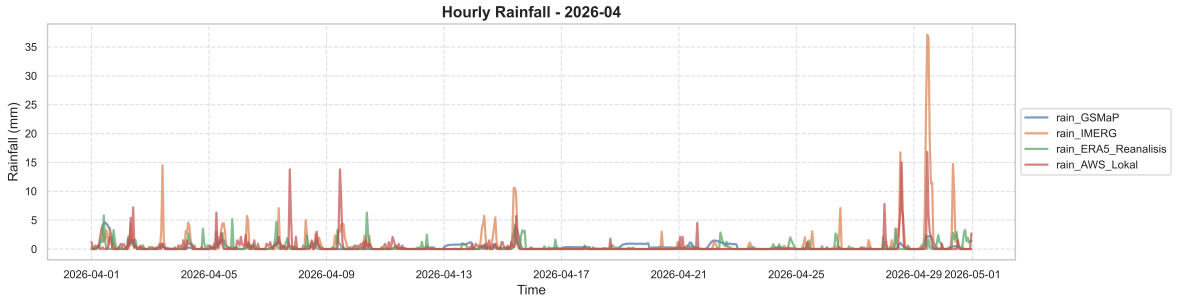

Hourly Rainfall - 2025-11
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal

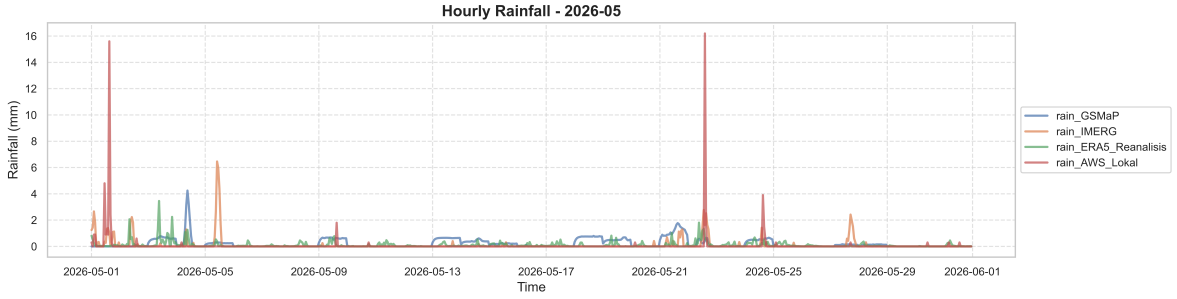

Hourly Rainfall - 2025-12
Rainfall (mm)
Time
rain_GSMaP
rain_IMERG
rain_ERA5_Reanalisis
rain_AWS_Lokal











5. Analisis Spesifik Curah Hujan

Analisis korelasi dan sebaran presipitasi membandingkan GPM IMERG, GSMaP, ERA5, dan AWS Lokal.

In [9]:
```
# Sel 11.1: Rain Correlation Setup & Scatter Plot Matrix
rain_cols = [c for c in df_merged.columns if c.startswith('rain_')]
if len(rain_cols) < 2:
    display(Markdown(" **Tidak cukup kolom curah hujan untuk perbandingan scatte
else:
    short_labels = {c: c.replace('rain_', '') for c in rain_cols}
```

```
    df_rain = df_merged[rain_cols].copy()
    df_rain.columns = [short_labels[c] for c in rain_cols]
    n_cols_rain = len(df_rain.columns)
    labels = list(df_rain.columns)
    display(Markdown(
f"##  Analisis Korelasi Scatterplot — Data Curah Hujan\n"
        f"Membandingkan **{n_cols_rain} sumber data curah hujan** secara pairwis
        "".join([f"- `{l}`\n" for l in labels])
    ))
    display(Markdown("### 1. Scatter Plot Matrix — Semua Pasangan Sumber Data"))
    cell_size = 3.5
    fig_size = cell_size * n_cols_rain
    fig, axes = plt.subplots(n_cols_rain, n_cols_rain,
                             figsize=(fig_size, fig_size),
                             squeeze=False)
    fig.suptitle('Scatter Plot Matrix — Korelasi Data Curah Hujan\n(Semua Pasang
                 fontsize=16, fontweight='bold', y=1.01)
    COLORS = plt.cm.tab10.colors
    for i, col_y in enumerate(labels):
        for j, col_x in enumerate(labels):
            ax = axes[i][j]
            if i == j:
                # KDE / Histogram
                data_diag = df_rain[col_y].dropna()
                ax.hist(data_diag, bins=50, color=COLORS[i % len(COLORS)],
                        alpha=0.7, edgecolor='white', linewidth=0.5)
                ax.set_facecolor('#f5f5f5')
                ax.set_title(col_y, fontsize=10, fontweight='bold', pad=6)
                ax.set_xlabel('mm/hr', fontsize=8)
                ax.set_ylabel('Freq', fontsize=8)
                ax.tick_params(labelsize=7)
            else:
                mask = df_rain[col_x].notna() & df_rain[col_y].notna()
                x_data = df_rain.loc[mask, col_x]
                y_data = df_rain.loc[mask, col_y]
                if len(x_data) < 5:
                    ax.text(0.5, 0.5, 'N/A', transform=ax.transAxes,
                            ha='center', va='center', fontsize=10, color='gray')
                    ax.set_visible(True)
                    continue
                # Pearson & Spearman
                r_pearson, p_pearson = stats.pearsonr(x_data, y_data)
                r_spearman, _ = stats.spearmanr(x_data, y_data)
                # Scatter
                n_plot = min(len(x_data), 5000)
                idx_sample = np.random.choice(len(x_data), n_plot, replace=False
                ax.scatter(
                    x_data.iloc[idx_sample],
                    y_data.iloc[idx_sample],
                    alpha=0.25, s=8,
                    color=COLORS[j % len(COLORS)],
                    edgecolors='none'
                )
                # Garis 1:1
                joint_min = max(0, min(x_data.min(), y_data.min()))
                joint_max = max(x_data.max(), y_data.max())
                ax.plot([joint_min, joint_max], [joint_min, joint_max],
                        'k--', linewidth=1.0, alpha=0.6, label='1:1')
                # Garis regresi linear
                slope, intercept, _, _, _ = stats.linregress(x_data, y_data)
```

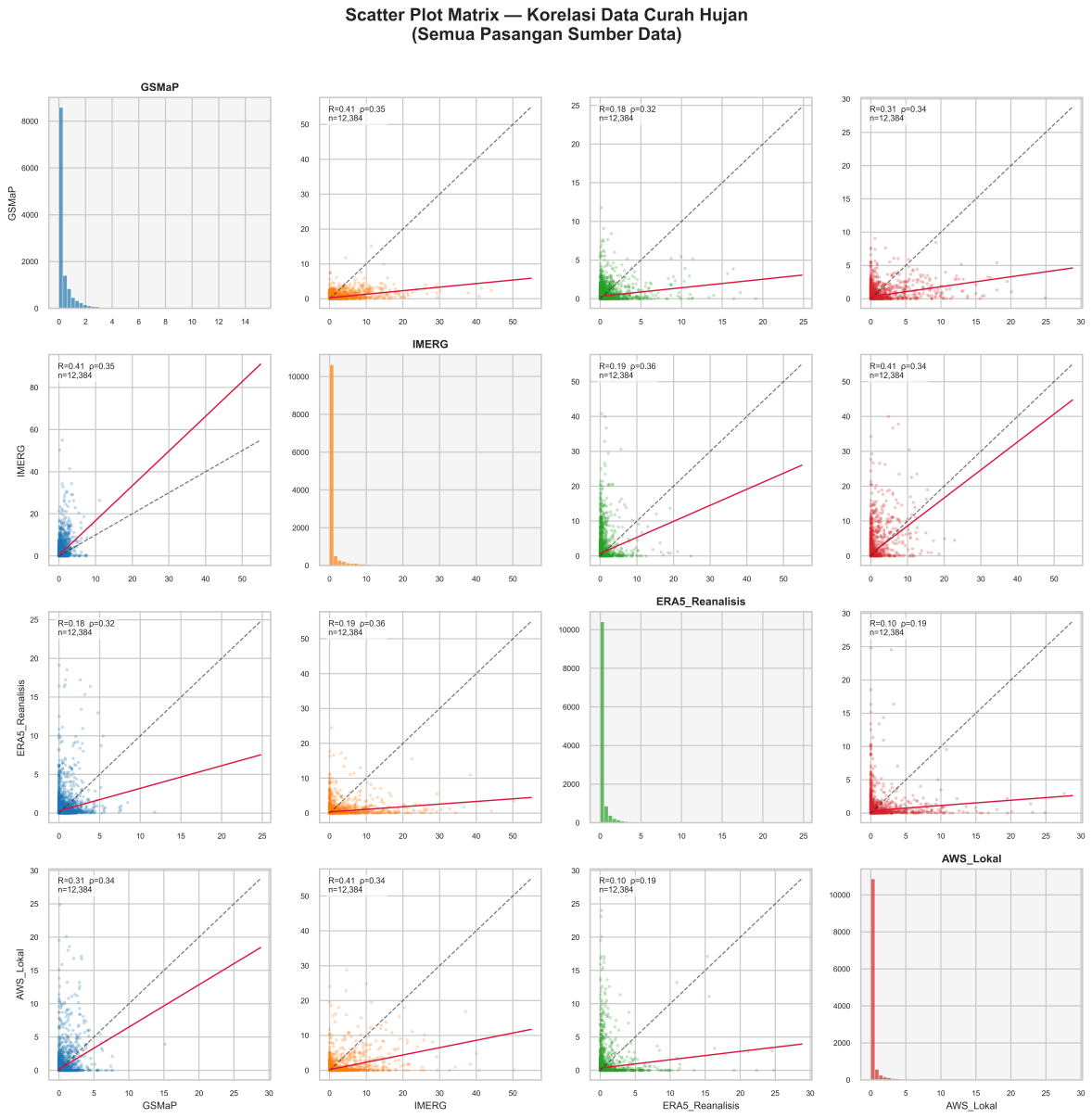
```
                x_line = np.linspace(joint_min, joint_max, 100)
                ax.plot(x_line, slope * x_line + intercept,
                        color='crimson', linewidth=1.2, label='OLS')
                # Annotasi R dan ρ
                txt = f"R={r_pearson:.2f}  ρ={r_spearman:.2f}\nn={len(x_data):,}
                ax.text(0.04, 0.96, txt, transform=ax.transAxes,
                        fontsize=7.5, va='top', ha='left',
                        bbox=dict(facecolor='white', alpha=0.7, edgecolor='none'
                ax.tick_params(labelsize=7)
            if i == n_cols_rain - 1:
                ax.set_xlabel(labels[j], fontsize=9, labelpad=4)
            else:
                ax.set_xlabel('')
            if j == 0:
                ax.set_ylabel(labels[i], fontsize=9, labelpad=4)
            else:
                ax.set_ylabel('')
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'rain_scatterplot_matrix.png'), dpi=600, b
    plt.show()
```

Analisis Korelasi Scatterplot — Data Curah Hujan

Membandingkan **4 sumber data curah hujan** secara pairwise:

- `GSMaP`
- `IMERG`
- `ERA5_Reanalisis`
- `AWS_Lokal`

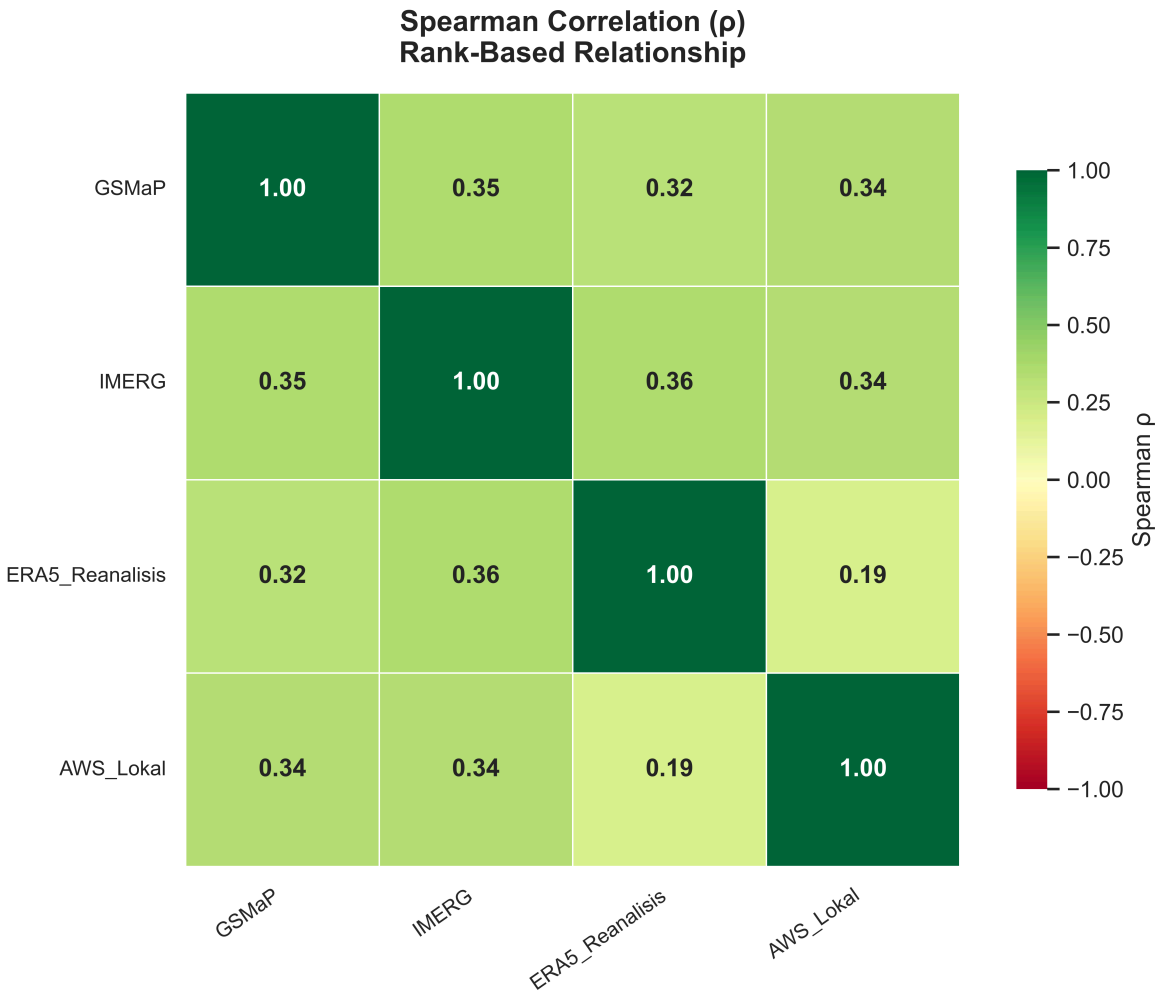
1. Scatter Plot Matrix — Semua Pasangan Sumber Data



In [10]:
```
## 5. Analisis Curah Hujan
```

In [11]:
```
# Sel 11.3: Heatmap Korelasi Spearman Curah Hujan
if 'df_rain' in locals() and not df_rain.empty:
    display(Markdown("### 2.2 Heatmap Korelasi Spearman — Curah Hujan"))
    corr_s = df_rain.corr(method='spearman').round(3)

    plt.figure(figsize=(9, 7))
    sns.heatmap(corr_s, annot=True, fmt='.2f', cmap='RdYlGn',
                vmin=-1, vmax=1, square=True,
                linewidths=0.5, linecolor='white',
                annot_kws={'size': 12, 'weight': 'bold'},
                cbar_kws={'label': 'Spearman ρ', 'shrink': 0.8})
    plt.title('Spearman Correlation (ρ)\nRank-Based Relationship', fontsize=14,
    plt.xticks(rotation=35, ha='right', fontsize=10)
    plt.yticks(rotation=0, fontsize=10)
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'rain_correlation_spearman_heatmap.png'),
    plt.show()
```

2.2 Heatmap Korelasi Spearman — Curah Hujan

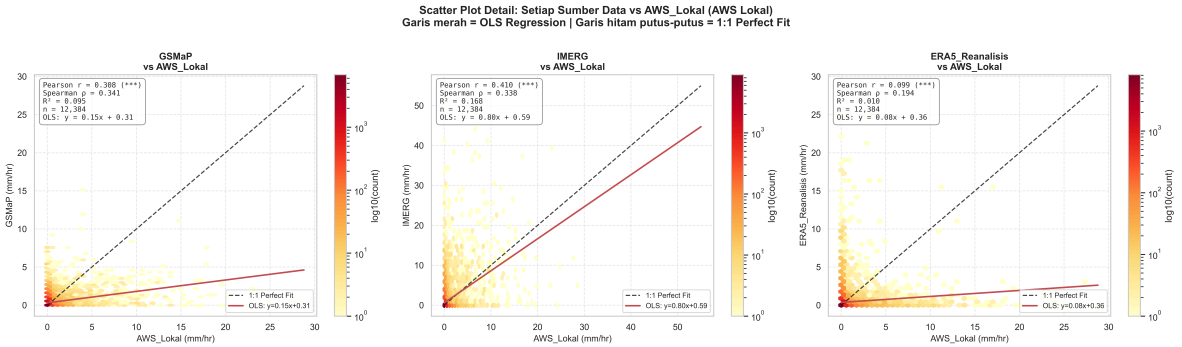


In [12]:

```
# Sel 11.4: Scatter Plot Detail vs AWS Lokal
if 'df_rain' in locals() and not df_rain.empty:
    display(Markdown("### 3. Scatter Plot Detail — Setiap Sumber Satelit vs Stas

    # Cari kolom AWS / IoT lokal
    aws_col = None
    for cand in ['AWS_Lokal', 'IoT', 'AWS', 'Lokal']:
        for lbl in labels:
            if cand.lower() in lbl.lower():
                aws_col = lbl
                break
        if aws_col:
            break
    if aws_col is None:
        display(Markdown("ℹ️ Tidak ada kolom AWS Lokal yang terdeteksi."))
    else:
        compare_labels = [l for l in labels if l != aws_col]
        n_compare = len(compare_labels)
        if n_compare > 0:
            ncols_sub = min(3, n_compare)
            nrows_sub = int(np.ceil(n_compare / ncols_sub))
            fig3, axes3 = plt.subplots(nrows_sub, ncols_sub,
                                       figsize=(7 * ncols_sub, 6 * nrows_sub),
                                       squeeze=False)
            fig3.suptitle(f'Scatter Plot Detail: Setiap Sumber Data vs {aws_col}
                          f'Garis merah = OLS Regression | Garis hitam putus-pu
                          fontsize=14, fontweight='bold', y=1.01)
            for idx, sat_col in enumerate(compare_labels):
                row_idx = idx // ncols_sub
```

```
            col_idx = idx % ncols_sub
            ax3 = axes3[row_idx][col_idx]
            mask = df_rain[aws_col].notna() & df_rain[sat_col].notna()
            x_aws = df_rain.loc[mask, aws_col]
            y_sat = df_rain.loc[mask, sat_col]
            if len(x_aws) < 5:
                ax3.text(0.5, 0.5, 'Tidak cukup data', transform=ax3.transAx
                         ha='center', va='center', fontsize=11, color='gray'
                ax3.set_title(sat_col, fontsize=12)
                continue
            r_p, p_val = stats.pearsonr(x_aws, y_aws := y_sat)
            r_s, _ = stats.spearmanr(x_aws, y_sat)
            slope, intercept, r_value, p_value, std_err = stats.linregress(x
            r2 = r_value ** 2
            # Subsample besar
            n_plot = min(len(x_aws), 8000)
            idx_s = np.random.choice(len(x_aws), n_plot, replace=False) if l
            # hexbin
            hb = ax3.hexbin(x_aws.values, y_sat.values,
                            gridsize=50, cmap='YlOrRd', mincnt=1, bins='log
            plt.colorbar(hb, ax=ax3, label='log10(count)')
            # Garis 1:1
            jmin = max(0, min(x_aws.min(), y_sat.min()))
            jmax = max(x_aws.max(), y_sat.max())
            ax3.plot([jmin, jmax], [jmin, jmax], 'k--', lw=1.5, alpha=0.8, l
            # Garis regresi
            x_line3 = np.linspace(jmin, jmax, 200)
            ax3.plot(x_line3, slope * x_line3 + intercept,
                     'r-', lw=2, label=f'OLS: y={slope:.2f}x+{intercept:.2f}
            # Statistik box
            sig_str = '***' if p_val < 0.001 else ('**' if p_val < 0.01 else
            stats_txt = (
                f"Pearson r = {r_p:.3f} ({sig_str})\n"
                f"Spearman ρ = {r_s:.3f}\n"
                f"R² = {r2:.3f}\n"
                f"n = {len(x_aws):,}\n"
                f"OLS: y = {slope:.2f}x + {intercept:.2f}"
            )
            ax3.text(0.03, 0.97, stats_txt, transform=ax3.transAxes,
                     fontsize=9, va='top', ha='left', family='monospace',
                     bbox=dict(facecolor='white', alpha=0.85, edgecolor='gra
                               boxstyle='round,pad=0.5'))
            ax3.set_xlabel(f'{aws_col} (mm/hr)', fontsize=11)
            ax3.set_ylabel(f'{sat_col} (mm/hr)', fontsize=11)
            ax3.set_title(f'{sat_col}\nvs {aws_col}', fontsize=12, fontweigh
            ax3.legend(fontsize=9, loc='lower right')
            ax3.grid(True, linestyle='--', alpha=0.4)
        for idx_empty in range(n_compare, nrows_sub * ncols_sub):
            axes3[idx_empty // ncols_sub][idx_empty % ncols_sub].set_visible
        plt.tight_layout()
        plt.savefig(os.path.join(out_dir, 'rain_scatter_details.png'), dpi=6
        plt.show()
```

3. Scatter Plot Detail — Setiap Sumber Satelit vs Stasiun AWS Lokal

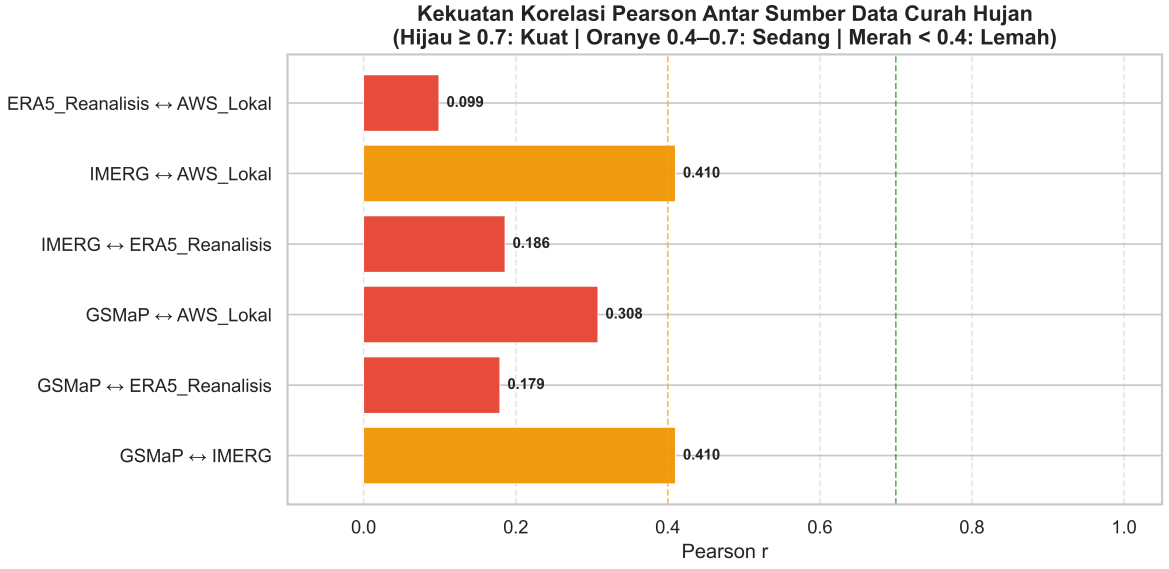


In [13]:
```
# Sel 11.5: Ringkasan Korelasi & Horizontal Bar Chart
if 'df_rain' in locals() and not df_rain.empty:
    display(Markdown("### 4. Ringkasan Statistik Korelasi Pairwise — Curah Hujan

    summary_rows = []
    for i_l, col_a in enumerate(labels):
        for j_l, col_b in enumerate(labels):
            if j_l <= i_l: continue
            mask = df_rain[col_a].notna() & df_rain[col_b].notna()
            xa = df_rain.loc[mask, col_a]
            xb = df_rain.loc[mask, col_b]
            if len(xa) < 5: continue

            r_p2, p_p2 = stats.pearsonr(xa, xb)
            r_s2, p_s2 = stats.spearmanr(xa, xb)
            slope2, intercept2, _, _, _ = stats.linregress(xa, xb)
            mbe2 = float(np.mean(xb - xa))
            mae2 = float(np.mean(np.abs(xb - xa)))
            rmse2 = float(np.sqrt(np.mean((xb - xa)**2)))

            summary_rows.append({
                'Pasangan': f'{col_a} ↔ {col_b}',
                'n': len(xa),
                'Pearson r': round(r_p2, 3),
                'Spearman ρ': round(r_s2, 3),
                'R²': round(r_p2**2, 3),
                'OLS slope': round(slope2, 3),
                'MBE (mm)': round(mbe2, 3),
                'MAE (mm)': round(mae2, 3),
                'RMSE (mm)': round(rmse2, 3),
            })
    if summary_rows:
        df_summary_rain = pd.DataFrame(summary_rows)
        display(df_summary_rain.sort_values('Pearson r', ascending=False).reset_
        # Bar chart Pearson r
        fig4, ax4 = plt.subplots(figsize=(max(10, len(summary_rows) * 1.5), 5))
        colors_bar = ['#2ecc71' if r >= 0.7 else '#f39c12' if r >= 0.4 else '#e7
                      for r in df_summary_rain['Pearson r']]
        bars = ax4.barh(df_summary_rain['Pasangan'], df_summary_rain['Pearson r'
                        color=colors_bar, edgecolor='white', linewidth=0.5)
        ax4.set_xlabel('Pearson r', fontsize=12)
        ax4.set_title('Kekuatan Korelasi Pearson Antar Sumber Data Curah Hujan\n
                      '(Hijau ≥ 0.7: Kuat | Oranye 0.4–0.7: Sedang | Merah < 0.
                      fontsize=13, fontweight='bold')
        ax4.set_xlim(-0.1, 1.05)
        ax4.axvline(x=0.7, color='green', linestyle='--', alpha=0.6, linewidth=1
        ax4.axvline(x=0.4, color='orange', linestyle='--', alpha=0.6, linewidth=
        for bar, val in zip(bars, df_summary_rain['Pearson r']):
            ax4.text(val + 0.01, bar.get_y() + bar.get_height()/2,
```

```
                    f'{val:.3f}', va='center', ha='left', fontsize=9, fontweigh
        ax4.grid(True, axis='x', linestyle='--', alpha=0.4)
        plt.tight_layout()
        plt.savefig(os.path.join(out_dir, 'rain_correlation_bar.png'), dpi=600,
        plt.show()
```

4. Ringkasan Statistik Korelasi Pairwise — Curah Hujan

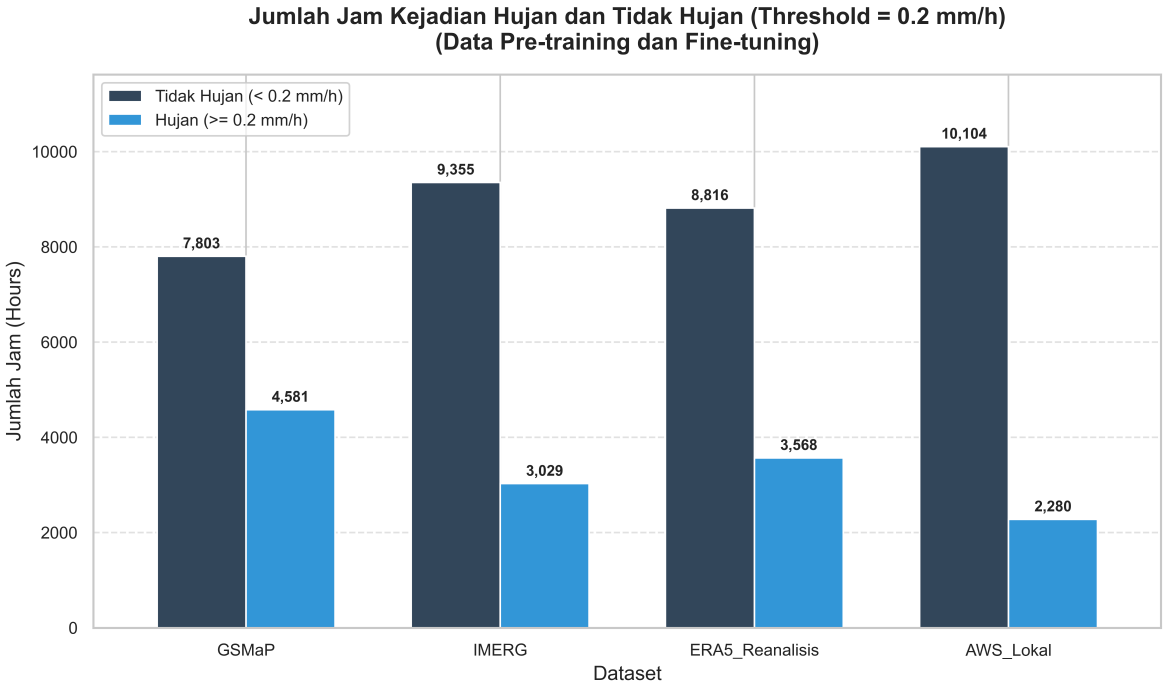
	Pasangan	n	Pearson r	Spearman ρ	R^2	OLS slope	MBE (mm)	MAE (mm)	RMSE (mm)
0	GSMaP ↔ IMERG	12384	0.410	0.354	0.168	1.650	0.530	0.917	2.855
1	IMERG ↔ AWS_Lokal	12384	0.410	0.338	0.168	0.210	-0.513	0.880	2.826
2	GSMaP ↔ AWS_Lokal	12384	0.308	0.341	0.095	0.635	0.017	0.525	1.500
3	IMERG ↔ ERA5_Reanalisis	12384	0.186	0.357	0.035	0.075	-0.506	1.001	3.091
4	GSMaP ↔ ERA5_Reanalisis	12384	0.179	0.316	0.032	0.292	0.024	0.515	1.320
5	ERA5_Reanalisis ↔ AWS_Lokal	12384	0.099	0.194	0.010	0.125	-0.007	0.628	1.879



In [14]:
```
# Sel 11.5: Visualisasi Jumlah Jam Hujan vs Tidak Hujan (Threshold = 0.2 mm/h)
if 'df_rain' in locals():

    thresh = 0.2

    # Hitung jumlah jam hujan dan tidak hujan
    data_counts = []
    for col in df_rain.columns:
        valid_data = df_rain[col].dropna()
        hujan = int(np.sum(valid_data >= thresh))
        tidak_hujan = int(np.sum(valid_data < thresh))
        data_counts.append({
            'Dataset': col,
            'Tidak Hujan (< 0.2 mm/h)': tidak_hujan,
            'Hujan (>= 0.2 mm/h)': hujan
```

```
        })

    df_counts = pd.DataFrame(data_counts)

    # Print tabel angka secara langsung
    print(df_counts.to_string(index=False))

    # Visualisasi Bar Chart Grouped
    df_plot = df_counts.set_index('Dataset')
    ax = df_plot.plot(kind='bar', figsize=(10, 6), color=['#34495e', '#3498db'],

    plt.title('Jumlah Jam Kejadian Hujan dan Tidak Hujan (Threshold = 0.2 mm/h)\
              '(Data Pre-training dan Fine-tuning)', fontsize=14, fontweight='bo
    plt.xlabel('Dataset', fontsize=12)
    plt.ylabel('Jumlah Jam (Hours)', fontsize=12)
    plt.xticks(rotation=0, fontsize=10)
    plt.yticks(fontsize=10)
    plt.legend(fontsize=10)
    plt.grid(True, axis='y', linestyle='--', alpha=0.5)

    # Tambahkan angka di atas setiap bar
    for p in ax.patches:
        height = p.get_height()
        ax.annotate(f'{int(height):,}',
                    xy=(p.get_x() + p.get_width() / 2, height),
                    xytext=(0, 3),
                    textcoords="offset points",
                    ha='center', va='bottom', fontsize=9, fontweight='bold')

    plt.ylim(0, max(df_counts['Tidak Hujan (< 0.2 mm/h)']) * 1.15)
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'rain_threshold_simulation_counts.png'), d
    plt.show()
else:
    print("DataFrame df_rain tidak ditemukan!")
```

```
       Dataset  Tidak Hujan (< 0.2 mm/h)  Hujan (>= 0.2 mm/h)
         GSMaP                      7803                 4581
         IMERG                      9355                 3029
ERA5_Reanalisis                     8816                 3568
     AWS_Lokal                     10104                 2280
```



In [15]:
```
audit_logs = []

missing = df_merged[[col for col in df_merged.columns if col.startswith('rain_')
negatives = (df_merged[[col for col in df_merged.columns if col.startswith('rain
duplicates = df_merged.index.duplicated().sum()

audit_table = pd.DataFrame({
    'Missing Values': missing,
    'Negative Values': negatives
})

display(Markdown(f"- **Duplicated Timestamps in Aligned Data**: {duplicates}\n")
display(audit_table)
```

- **Duplicated Timestamps in Aligned Data**: 0

	Missing Values	Negative Values
rain_GSMaP	0	0
rain_IMERG	0	0
rain_ERA5_Reanalisis	0	0
rain_AWS_Lokal	0	0

6. Ringkasan Evaluasi Data Satelit vs AWS Lokal

- **GPM IMERG & GSMaP:** Efektif menangkap dinamika temporal dan sebaran presipitasi lokal.
- **ERA5 Reanalisis:** Akurat untuk parameter suhu, tekanan, dan kelembapan.

7. Ekspor Data Hasil Analisis

Penyimpanan hasil penggabungan data dan evaluasi ke berkas CSV.

In [16]:
```
## 6. Ringkasan Evaluasi
```

In [17]:
```
## 7. Ekspor Data
```

8. Uji Signifikansi Statistik Curah Hujan

Evaluasi perbedaan distribusi dan median curah hujan menggunakan **Uji Wilcoxon Signed-Rank** (paired) dan **Uji Mann-Whitney U** (musim hujan vs kemarau).

In [18]:
```
# Sel 11.7: Uji Wilcoxon Signed-Rank & Mann-Whitney U untuk Curah Hujan

# 1. Uji Wilcoxon Signed-Rank (Data Berpasangan/Paired)
# Membandingkan median curah hujan pada jam yang sama (hanya membandingkan jam s
pairs = [
    ('rain_IMERG', 'rain_ERA5_Reanalisis', 'IMERG vs ERA5 Reanalisis'),
    ('rain_GSMaP', 'rain_ERA5_Reanalisis', 'GSMaP vs ERA5 Reanalisis'),
    ('rain_IMERG', 'rain_AWS_Lokal', 'IMERG vs AWS Lokal'),
    ('rain_GSMaP', 'rain_AWS_Lokal', 'GSMaP vs AWS Lokal'),
    ('rain_ERA5_Reanalisis', 'rain_AWS_Lokal', 'ERA5 Reanalisis vs AWS Lokal')
]

wilcoxon_results = []
for col1, col2, label in pairs:
    if col1 in df_merged.columns and col2 in df_merged.columns:
        # Drop NaNs and filter for wet hours
        df_temp = df_merged[[col1, col2]].dropna()
        df_wet = df_temp[(df_temp[col1] > 0.1) | (df_temp[col2] > 0.1)]

        if len(df_wet) > 10:
            stat, p_val = stats.wilcoxon(df_wet[col1], df_wet[col2])
            significant = "Ya (p < 0.05)" if p_val < 0.05 else "Tidak (p >= 0.05
            wilcoxon_results.append({
                'Perbandingan': label,
                'Jumlah Sampel': len(df_wet),
                'Statistik Uji': round(stat, 2),
                'p-value': f"{p_val:.4e}",
                'Berbeda Nyata?': significant
            })

df_wilcoxon = pd.DataFrame(wilcoxon_results)
display(Markdown("### Hasil Uji Wilcoxon Signed-Rank (Paired)"))
display(df_wilcoxon)

# 2. Uji Mann-Whitney U (Data Tidak Berpasangan/Unpaired)
# Membandingkan distribusi curah hujan antara Musim Hujan (Nov-Apr) vs Musim Kem
df_merged['month'] = df_merged.index.month
wet_months = [11, 12, 1, 2, 3, 4]
dry_months = [5, 6, 7, 8, 9, 10]

mw_results = []
datasets = [
    ('rain_IMERG', 'GPM IMERG'),
    ('rain_GSMaP', 'GSMaP'),
    ('rain_ERA5_Reanalisis', 'ERA5 Reanalisis'),
    ('rain_AWS_Lokal', 'AWS Lokal')
```

```
]

for col, label in datasets:
    if col in df_merged.columns:
        wet_data = df_merged[df_merged['month'].isin(wet_months)][col].dropna()
        dry_data = df_merged[df_merged['month'].isin(dry_months)][col].dropna()

        if len(wet_data) > 10 and len(dry_data) > 10:
            stat, p_val = stats.mannwhitneyu(wet_data, dry_data, alternative='tw
            significant = "Ya (p < 0.05)" if p_val < 0.05 else "Tidak (p >= 0.05
            mw_results.append({
                'Dataset': label,
                'Sampel Musim Hujan': len(wet_data),
                'Sampel Musim Kemarau': len(dry_data),
                'Statistik Uji': round(stat, 2),
                'p-value': f"{p_val:.4e}",
                'Perbedaan Signifikan?': significant
            })

df_mw = pd.DataFrame(mw_results)
display(Markdown("### Hasil Uji Mann-Whitney U (Unpaired: Musim Hujan vs Musim K
display(df_mw)
```

Hasil Uji Wilcoxon Signed-Rank (Paired)

	Perbandingan	Jumlah Sampel	Statistik Uji	p-value	Berbeda Nyata?
0	IMERG vs ERA5 Reanalisis	6242	8600403.0	1.0626e-15	Ya (p < 0.05)
1	GSMaP vs ERA5 Reanalisis	7630	12256530.0	6.3944e-33	Ya (p < 0.05)
2	IMERG vs AWS Lokal	4543	3065131.0	3.1727e-124	Ya (p < 0.05)
3	GSMaP vs AWS Lokal	6258	6402977.0	2.6437e-124	Ya (p < 0.05)
4	ERA5 Reanalisis vs AWS Lokal	6031	7118459.0	2.2488e-48	Ya (p < 0.05)

Hasil Uji Mann-Whitney U (Unpaired: Musim Hujan vs Musim Kemarau)

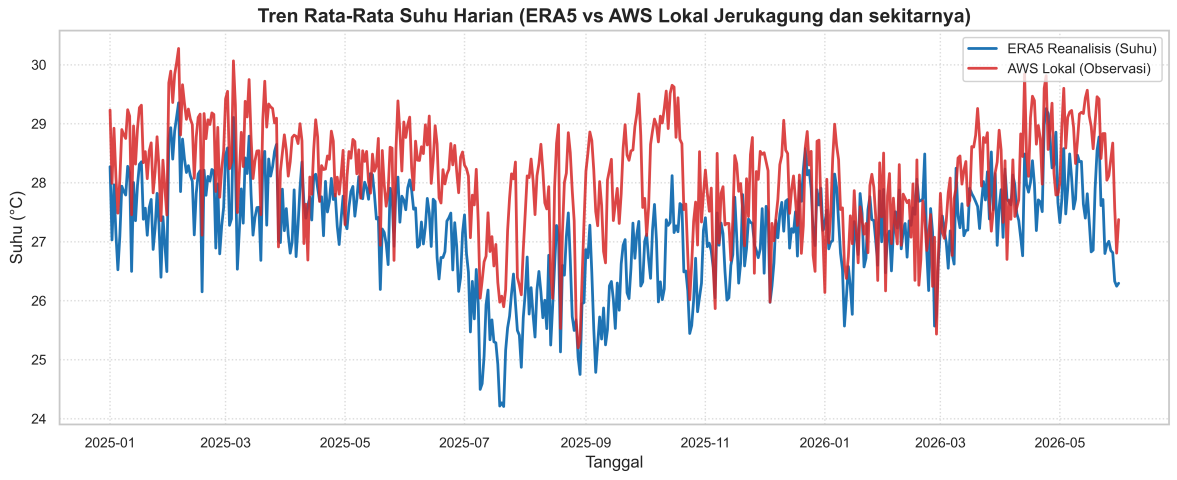
	Dataset	Sampel Musim Hujan	Sampel Musim Kemarau	Statistik Uji	p-value	Perbedaan Signifikan?
0	GPM IMERG	7224	5160	22658951.0	7.2312e-137	Ya (p < 0.05)
1	GSMaP	7224	5160	19528334.0	7.1302e-07	Ya (p < 0.05)
2	ERA5 Reanalisis	7224	5160	20648724.5	1.0168e-24	Ya (p < 0.05)
3	AWS Lokal	7224	5160	22845715.0	1.9617e-199	Ya (p < 0.05)

8. Uji Signifikansi Statistik

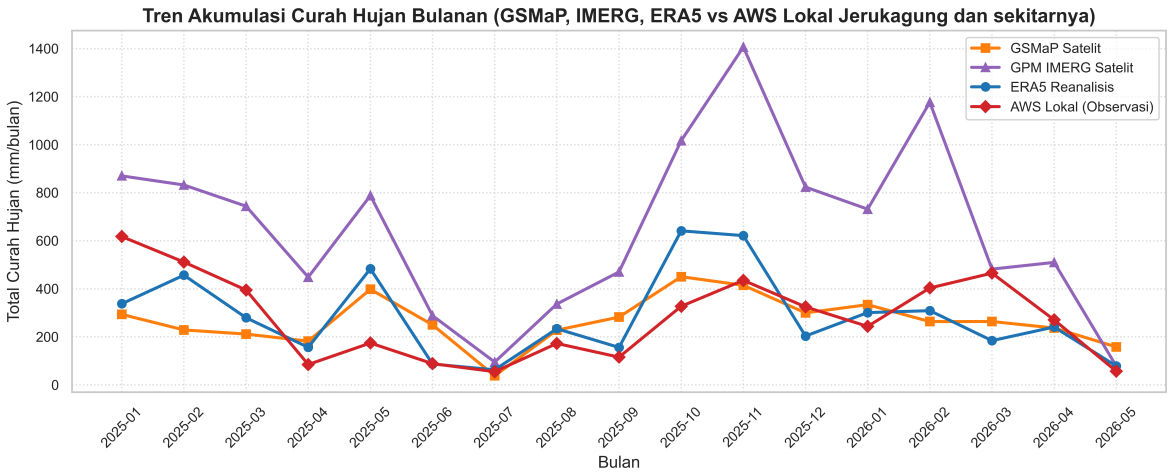
In [19]:
```
# Sel 11.8: Line Chart Analisis Tren Suhu dan Curah Hujan (ERA5, GSMaP, IMERG, A

# 1. Tren Rata-Rata Suhu Harian (ERA5 vs AWS Lokal)
temp_cols = [c for c in ['temperature_ERA5_Reanalisis', 'temperature_AWS_Lokal']
if len(temp_cols) > 0:
    df_daily_temp = df_merged[temp_cols].resample('D').mean()

    plt.figure(figsize=(12, 5))
    if 'temperature_ERA5_Reanalisis' in df_daily_temp.columns:
        plt.plot(df_daily_temp.index, df_daily_temp['temperature_ERA5_Reanalisis
    if 'temperature_AWS_Lokal' in df_daily_temp.columns:
        plt.plot(df_daily_temp.index, df_daily_temp['temperature_AWS_Lokal'], la

    plt.title('Tren Rata-Rata Suhu Harian (ERA5 vs AWS Lokal Jerukagung dan seki
    plt.xlabel('Tanggal', fontsize=12)
    plt.ylabel('Suhu (°C)', fontsize=12)
    plt.xticks(fontsize=10)
    plt.yticks(fontsize=10)
    plt.legend(loc='upper right', fontsize=10)
    plt.grid(True, linestyle=':', alpha=0.6)
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'trend_linechart_temperature_daily.png'),
    display(Markdown("### 1. Tren Rata-Rata Suhu Harian (ERA5 vs AWS Lokal)"))
    plt.show()

# 2. Tren Akumulasi Curah Hujan Bulanan (GSMaP, IMERG, ERA5 vs AWS Lokal)
rain_cols = ['rain_GSMaP', 'rain_IMERG', 'rain_ERA5_Reanalisis', 'rain_AWS_Lokal
avail_rain_cols = [c for c in rain_cols if c in df_merged.columns]

if len(avail_rain_cols) > 0:
    df_monthly_rain = df_merged[avail_rain_cols].resample('MS').sum()

    colors = {'rain_GSMaP': '#ff7f0e', 'rain_IMERG': '#9467bd', 'rain_ERA5_Reana
    labels = {'rain_GSMaP': 'GSMaP Satelit', 'rain_IMERG': 'GPM IMERG Satelit',
    markers = {'rain_GSMaP': 's', 'rain_IMERG': '^', 'rain_ERA5_Reanalisis': 'o'

    plt.figure(figsize=(12, 5))
    for col in avail_rain_cols:
        plt.plot(
            df_monthly_rain.index.strftime('%Y-%m'),
            df_monthly_rain[col],
            label=labels.get(col, col),
            color=colors.get(col, '#333333'),
            linewidth=2.2,
            marker=markers.get(col, 'o'),
            markersize=6
        )

    plt.title('Tren Akumulasi Curah Hujan Bulanan (GSMaP, IMERG, ERA5 vs AWS Lok
    plt.xlabel('Bulan', fontsize=12)
    plt.ylabel('Total Curah Hujan (mm/bulan)', fontsize=12)
    plt.xticks(rotation=45, fontsize=10)
    plt.yticks(fontsize=10)
    plt.legend(loc='upper right', fontsize=10)
    plt.grid(True, linestyle=':', alpha=0.6)
    plt.tight_layout()
```

```
    plt.savefig(os.path.join(out_dir, 'trend_linechart_rainfall_monthly.png'), d
    display(Markdown("### 2. Tren Akumulasi Curah Hujan Bulanan (GSMaP, IMERG, E
    plt.show()

    # 3. Tren Curah Hujan Rata-Rata 7-Hari (Moving Average)
    df_daily_rain = df_merged[avail_rain_cols].resample('D').sum()
    df_rolling_rain = df_daily_rain.rolling(window=7, min_periods=1).mean()

    plt.figure(figsize=(12, 5))
    for col in avail_rain_cols:
        plt.plot(
            df_rolling_rain.index,
            df_rolling_rain[col],
            label=labels.get(col, col),
            color=colors.get(col, '#333333'),
            linewidth=2.0
        )

    plt.title('Tren Akumulasi Curah Hujan Harian Rata-Rata 7-Hari (GSMaP, IMERG,
    plt.xlabel('Tanggal', fontsize=12)
    plt.ylabel('Rerata Hujan 7-Hari (mm/hari)', fontsize=12)
    plt.xticks(fontsize=10)
    plt.yticks(fontsize=10)
    plt.legend(loc='upper right', fontsize=10)
    plt.grid(True, linestyle=':', alpha=0.6)
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'trend_linechart_rainfall_7day_moving.png'
    display(Markdown("### 3. Tren Curah Hujan Rata-Rata 7-Hari (GSMaP, IMERG, ER
    plt.show()
```

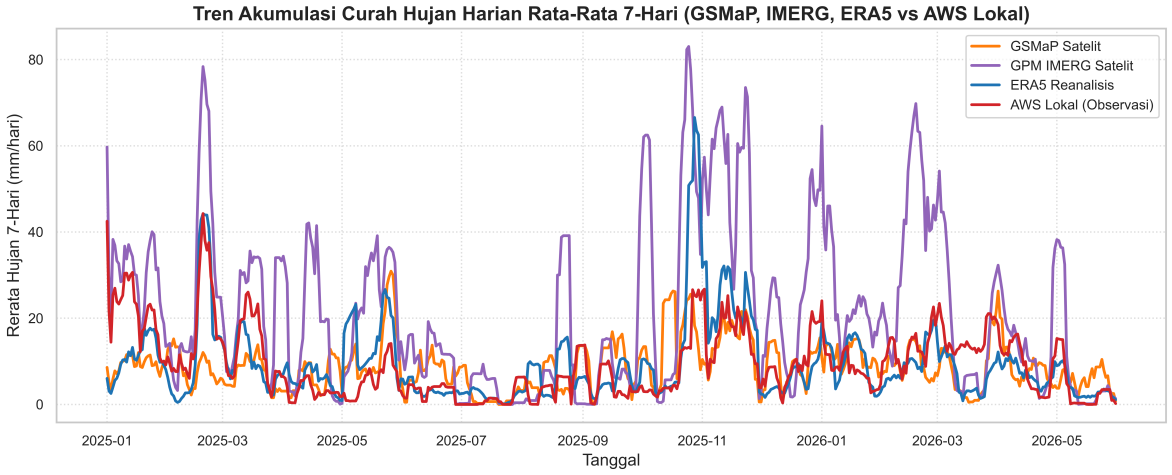
1. Tren Rata-Rata Suhu Harian (ERA5 vs AWS Lokal)



2. Tren Akumulasi Curah Hujan Bulanan (GSMaP, IMERG, ERA5, AWS Lokal)



3. Tren Curah Hujan Rata-Rata 7-Hari (GSMaP, IMERG, ERA5, AWS Lokal)



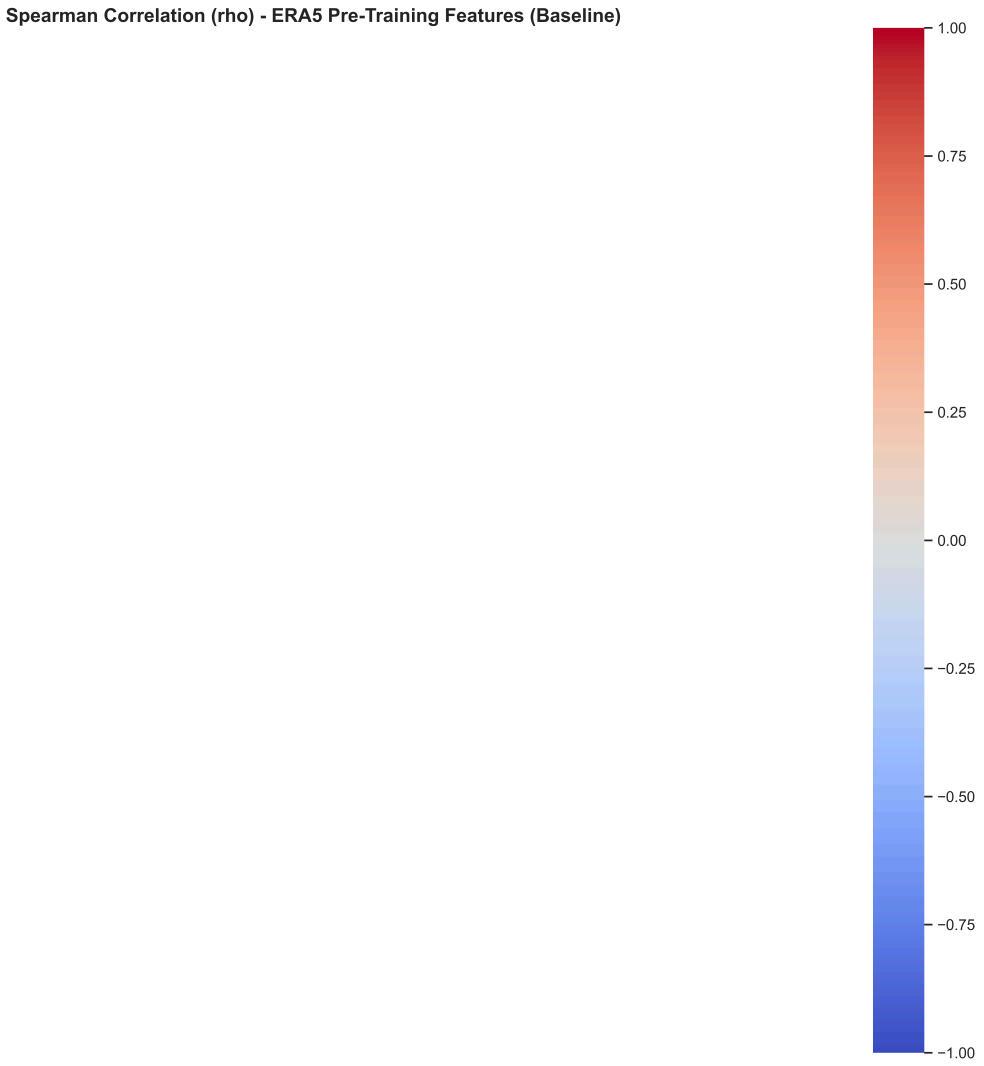
9. Tren Temporal

In [20]:
```
# Sel 12: Analisis Korelasi Pearson & Spearman Fitur Pre-Training ERA5 (Baseline
ERA5_FEATURES_MAPPING = {
    'temperature_era5': 'temperature',
    'humidity_era5': 'humidity',
    'dewpoint_era5': 'dewpoint',
    'rain_mm': 'rainrate',
    'sealevel_pressure_era5': 'pressure',
    'u10_era5': 'era5_u_wind',
    'v10_era5': 'era5_v_wind',
    'cloud_cover_era5': 'era5_cloud_cover',
    'cape_era5': 'era5_cape',
    'total_column_water_vapour_era5': 'era5_tcwv',
    'moisture_divergence_era5': 'era5_moisture_div',
    'direct_radiation_era5': 'era5_direct_rad',
    'sunshine_duration_era5': 'era5_sunshine'
}
if 'datasets_raw' in locals() and 'ERA5_Reanalisis' in datasets_raw:
    df_raw = datasets_raw['ERA5_Reanalisis'].copy()

    # Parse waktu
    if "datetime_utc" in df_raw.columns:
        df_raw["timestamp"] = pd.to_datetime(df_raw["datetime_utc"], utc=True)
    else:
```

```
        df_raw["timestamp"] = pd.to_datetime(df_raw["unixtime"], unit="s", utc=T

    df_raw = df_raw.set_index("timestamp").sort_index()

    # Filter sesuai timeline evaluasi
    df_filtered = df_raw[(df_raw.index >= '2025-01-01 00:00:00') & (df_raw.index

    # Ambil kolom yang terdaftar di mapping
    existing_cols = [c for c in ERA5_FEATURES_MAPPING.keys() if c in df_filtered
    df_features = df_filtered[existing_cols].copy()

    # Rename ke nama mapped pre-training
    df_features = df_features.rename(columns=ERA5_FEATURES_MAPPING)

    # Hitung Korelasi
    corr_p_feat = df_features.corr(method='pearson')
    corr_s_feat = df_features.corr(method='spearman')

    # 1. Pearson Heatmap
    display(Markdown("### Matriks Korelasi Pearson Fitur Pre-Training ERA5 (Base
    plt.figure(figsize=(13, 11))
    sns.heatmap(corr_p_feat, annot=True, fmt='.2f', cmap='coolwarm',
                vmin=-1, vmax=1, square=True, linewidths=0.5, linecolor='white',
                annot_kws={'size': 10, 'weight': 'bold'})
    plt.title('Pearson Correlation (r) - ERA5 Pre-Training Features (Baseline)',
    plt.xticks(rotation=45, ha='right', fontsize=10)
    plt.yticks(rotation=0, fontsize=10)
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'pretraining_features_pearson_heatmap.png'
    plt.show()

    # 2. Spearman Heatmap
    display(Markdown("### Matriks Korelasi Spearman Fitur Pre-Training ERA5 (Bas
    plt.figure(figsize=(13, 11))
    sns.heatmap(corr_s_feat, annot=True, fmt='.2f', cmap='coolwarm',
                vmin=-1, vmax=1, square=True, linewidths=0.5, linecolor='white',
                annot_kws={'size': 10, 'weight': 'bold'})
    plt.title('Spearman Correlation (rho) - ERA5 Pre-Training Features (Baseline
    plt.xticks(rotation=45, ha='right', fontsize=10)
    plt.yticks(rotation=0, fontsize=10)
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'pretraining_features_spearman_heatmap.png
    plt.show()
else:
    print("Dataset ERA5_Reanalisis tidak ditemukan di datasets_raw!")
```

Matriks Korelasi Pearson Fitur Pre-Training ERA5 (Baseline)

Matriks Korelasi Spearman Fitur Pre-Training ERA5 (Baseline)

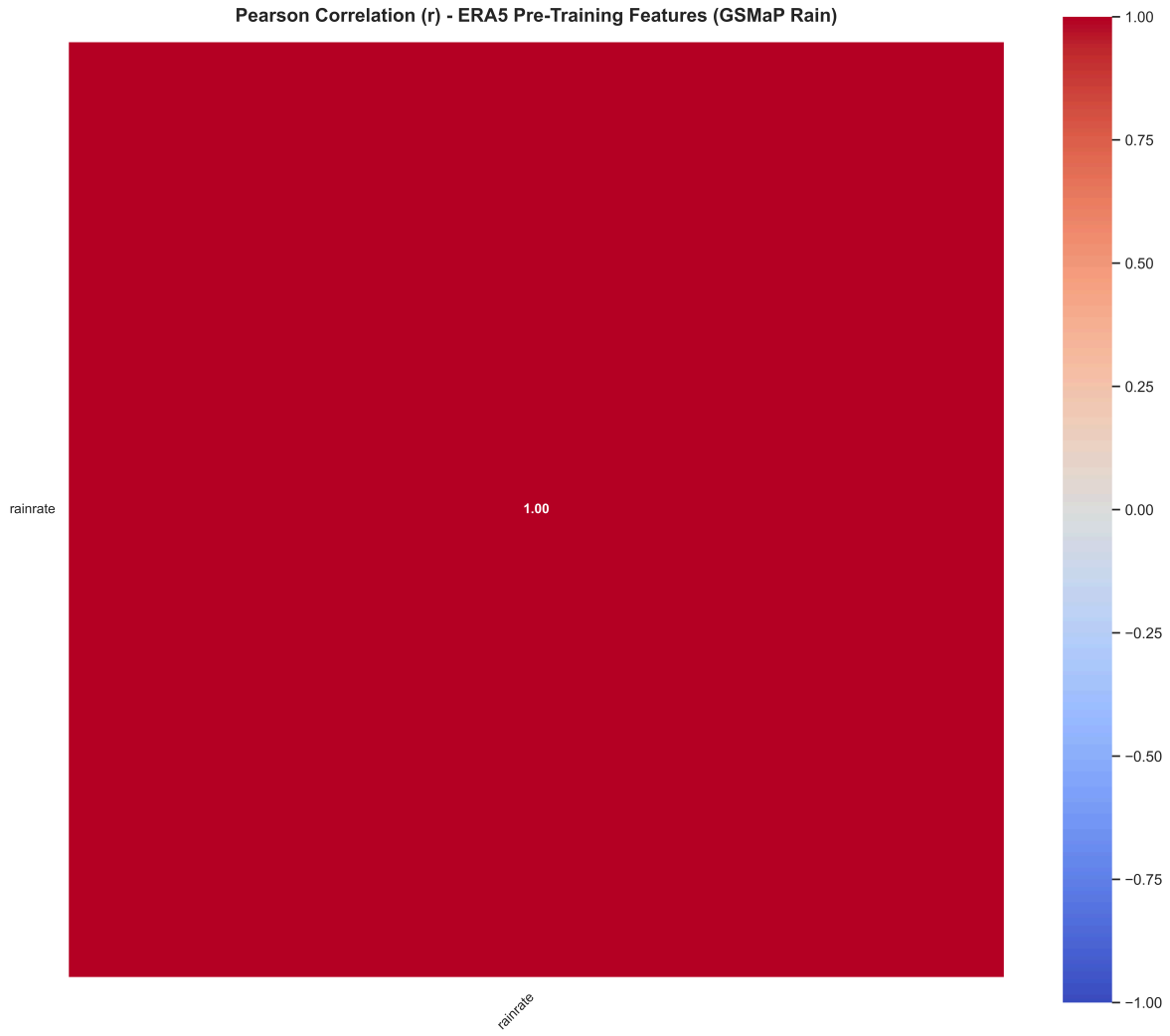


In [21]:
```
## 10. Korelasi Fitur Pre-Training ERA5
```

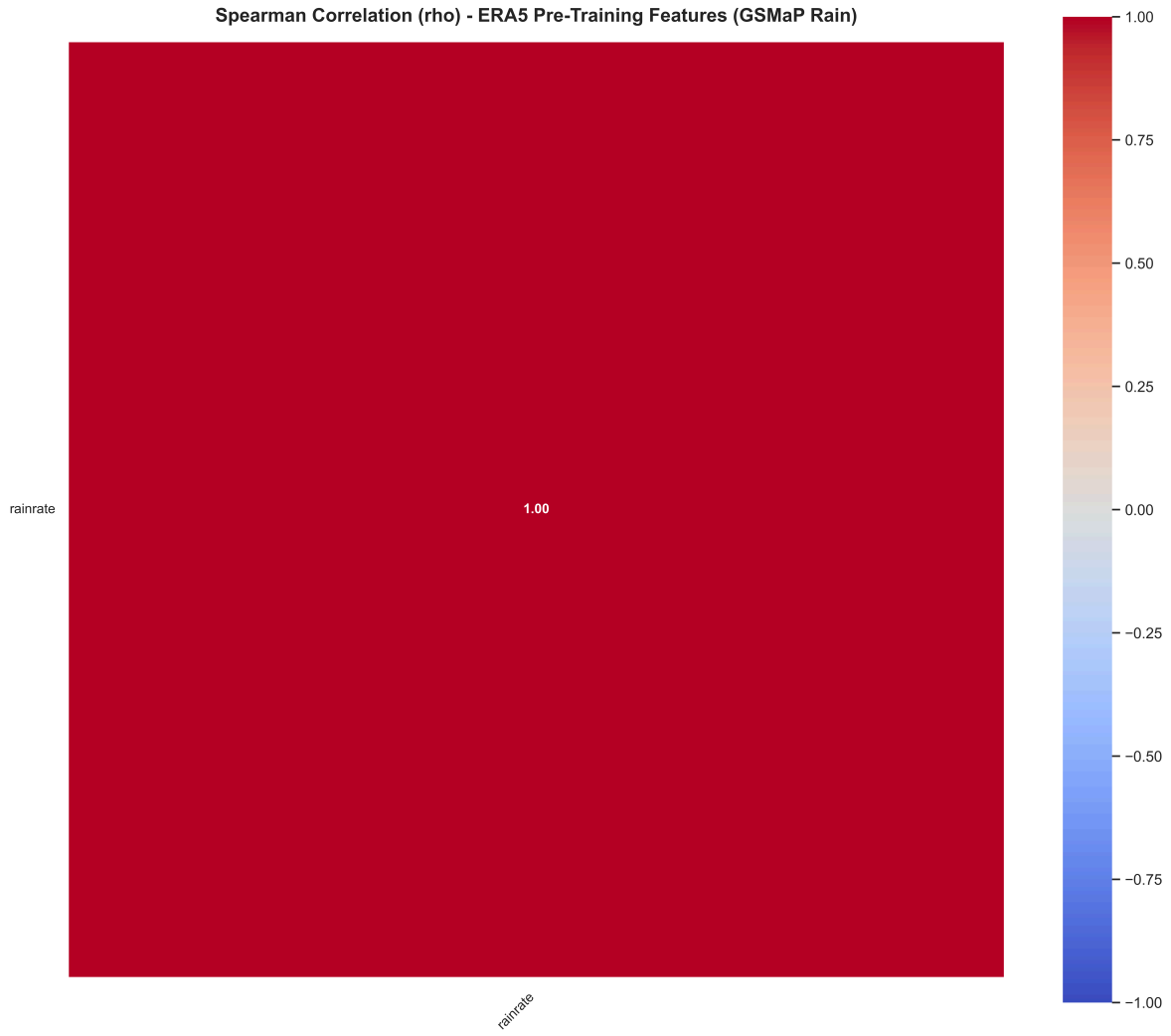
In [22]:
```
# Sel 14: Analisis Korelasi Pearson & Spearman Fitur Pre-Training ERA5 (Rain dig
if 'datasets_raw' in locals() and 'ERA5_Reanalisis' in datasets_raw and 'df_merg
    df_raw = datasets_raw['ERA5_Reanalisis'].copy()

    # Parse waktu
    if "datetime_utc" in df_raw.columns:
        df_raw["timestamp"] = pd.to_datetime(df_raw["datetime_utc"], utc=True)
    else:
        df_raw["timestamp"] = pd.to_datetime(df_raw["unixtime"], unit="s", utc=T

    df_raw = df_raw.set_index("timestamp").sort_index()

    # Filter sesuai timeline evaluasi
    df_filtered = df_raw[(df_raw.index >= '2025-01-01 00:00:00') & (df_raw.index

    # Ambil kolom yang terdaftar di mapping
    existing_cols = [c for c in ERA5_FEATURES_MAPPING.keys() if c in df_filtered
    df_features = df_filtered[existing_cols].copy()

    # Rename ke nama mapped pre-training
    df_features = df_features.rename(columns=ERA5_FEATURES_MAPPING)

    # Alinyemen dan ganti kolom rainrate dengan rain_GSMaP dari df_merged
    df_features['rainrate'] = df_merged['rain_GSMaP'].reindex(df_features.index)
```

```
    # Hitung Korelasi
    corr_p_feat = df_features.corr(method='pearson')
    corr_s_feat = df_features.corr(method='spearman')

    # 1. Pearson Heatmap
    display(Markdown("### Matriks Korelasi Pearson Fitur Pre-Training ERA5 (Rain
    plt.figure(figsize=(13, 11))
    sns.heatmap(corr_p_feat, annot=True, fmt='.2f', cmap='coolwarm',
                vmin=-1, vmax=1, square=True, linewidths=0.5, linecolor='white',
                annot_kws={'size': 10, 'weight': 'bold'})
    plt.title('Pearson Correlation (r) - ERA5 Pre-Training Features (GSMaP Rain)
    plt.xticks(rotation=45, ha='right', fontsize=10)
    plt.yticks(rotation=0, fontsize=10)
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'pretraining_features_pearson_heatmap_gsma
    plt.show()

    # 2. Spearman Heatmap
    display(Markdown("### Matriks Korelasi Spearman Fitur Pre-Training ERA5 (Rai
    plt.figure(figsize=(13, 11))
    sns.heatmap(corr_s_feat, annot=True, fmt='.2f', cmap='coolwarm',
                vmin=-1, vmax=1, square=True, linewidths=0.5, linecolor='white',
                annot_kws={'size': 10, 'weight': 'bold'})
    plt.title('Spearman Correlation (rho) - ERA5 Pre-Training Features (GSMaP Ra
    plt.xticks(rotation=45, ha='right', fontsize=10)
    plt.yticks(rotation=0, fontsize=10)
    plt.tight_layout()
    plt.savefig(os.path.join(out_dir, 'pretraining_features_spearman_heatmap_gsm
    plt.show()
else:
    print("Dataset atau df_merged tidak tersedia!")
```

Matriks Korelasi Pearson Fitur Pre-Training ERA5 (Rain diganti GSMaP)



Matriks Korelasi Spearman Fitur Pre-Training ERA5 (Rain diganti GSMaP)



11. Analisis Tren Iklim Jangka Panjang (2005 – 2025)

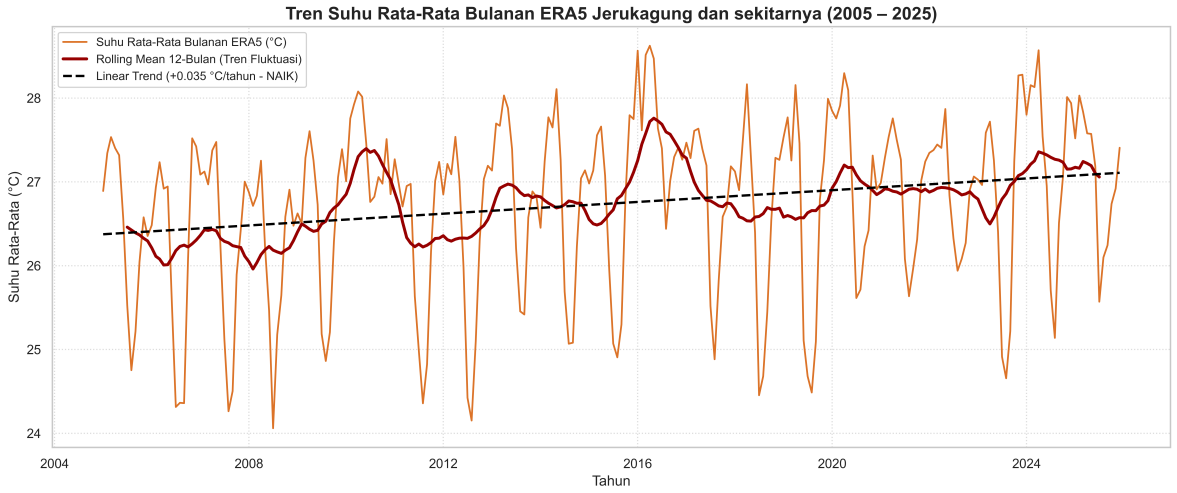
Analisis tren 21 tahun untuk suhu (ERA5) dan curah hujan (GPM IMERG, GSMaP, ERA5) di Jerukagung dan sekitarnya.

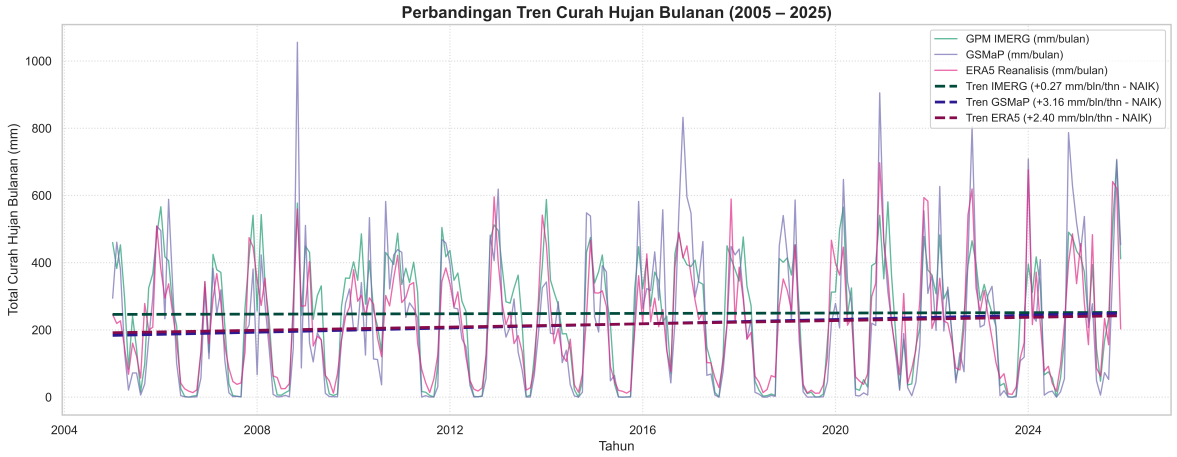
In [23]:

```
# Sel 15.1: Analisis Linechart Tren Bulanan Suhu & Curah Hujan (2005 - 2025)
base_sat_dir = os.path.join(base_dir, 'Data_Satelit') if 'base_dir' in locals()
out_dir = os.path.join(base_dir, 'Hasil_Analisis') if 'base_dir' in locals() els
os.makedirs(out_dir, exist_ok=True)
# 1. Load Long-Term Datasets (2005 - 2025)
print('Memuat data jangka panjang 2005 - 2025...')
df_era5_lt = pd.read_csv(os.path.join(base_sat_dir, 'ERA5_Hourly_All_Requested_F
df_era5_lt['datetime_utc'] = pd.to_datetime(df_era5_lt['datetime_utc'])
df_era5_lt.set_index('datetime_utc', inplace=True)
df_era5_lt = df_era5_lt['2005-01-01':'2025-12-31']
if df_era5_lt['temperature'].mean() > 200:
    df_era5_lt['temperature_C'] = df_era5_lt['temperature'] - 273.15
else:
    df_era5_lt['temperature_C'] = df_era5_lt['temperature']
df_gsmap_lt = pd.read_csv(os.path.join(base_sat_dir, 'Rainfall_GSMap_TimeSeries_
df_gsmap_lt['datetime_utc'] = pd.to_datetime(df_gsmap_lt['datetime_utc'])
df_gsmap_lt.set_index('datetime_utc', inplace=True)
df_gsmap_lt = df_gsmap_lt['2005-01-01':'2025-12-31']
df_imerg_lt = pd.read_csv(os.path.join(base_sat_dir, 'Rainfall_IMERG_TimeSeries_
```

```
df_imerg_lt['datetime_utc'] = pd.to_datetime(df_imerg_lt['datetime_utc'])
df_imerg_lt.set_index('datetime_utc', inplace=True)
df_imerg_lt = df_imerg_lt['2005-01-01':'2025-12-31']
df_imerg_hourly_lt = df_imerg_lt.resample('1h').mean()
# Aggregasi Bulanan (2005 - 2025)
monthly_temp = df_era5_lt['temperature_C'].resample('1MS').mean()
monthly_rain_era5 = df_era5_lt['rainrate'].resample('1MS').sum()
monthly_rain_gsmap = df_gsmap_lt['hourlyPrecipRate'].resample('1MS').sum()
monthly_rain_imerg = df_imerg_hourly_lt['precipitation'].resample('1MS').sum()
# ─── 1. LINECHART TREN SUHU BULANAN (ERA5 2005 - 2025) ───
plt.figure(figsize=(14, 6))
plt.plot(monthly_temp.index, monthly_temp.values, color='#d95f02', linewidth=1.5
plt.plot(monthly_temp.index, monthly_temp.rolling(12, center=True).mean(), color
# Fit linear trendline suhu
x_numeric = (monthly_temp.index - monthly_temp.index[0]).total_seconds() / (8640
slope_t, intercept_t = np.polyfit(x_numeric, monthly_temp.values, 1)
trend_line_t = slope_t * x_numeric + intercept_t
plt.plot(monthly_temp.index, trend_line_t, color='black', linestyle='--', linewi
plt.title('Tren Suhu Rata-Rata Bulanan ERA5 Jerukagung dan sekitarnya (2005 – 20
plt.xlabel('Tahun', fontsize=12)
plt.ylabel('Suhu Rata-Rata (°C)', fontsize=12)
plt.grid(True, linestyle=':', alpha=0.6)
plt.legend(loc='upper left', fontsize=10)
plt.tight_layout()
plt.savefig(os.path.join(out_dir, 'monthly_temperature_trend_2005_2025.png'), dp
plt.show()
# ─── 2. LINECHART TREN CURAH HUJAN BULANAN (GSMaP, IMERG, ERA5 2005 - 2025) DEN
plt.figure(figsize=(15, 6))
plt.plot(monthly_rain_imerg.index, monthly_rain_imerg.values, color='#1b9e77', l
plt.plot(monthly_rain_gsmap.index, monthly_rain_gsmap.values, color='#7570b3', l
plt.plot(monthly_rain_era5.index, monthly_rain_era5.values, color='#e7298a', lin
# Hitung Garis Tren Linier untuk Masing-masing Dataset Presipitasi
slope_imerg, int_imerg = np.polyfit(x_numeric, monthly_rain_imerg.values, 1)
slope_gsmap, int_gsmap = np.polyfit(x_numeric, monthly_rain_gsmap.values, 1)
slope_era5, int_era5 = np.polyfit(x_numeric, monthly_rain_era5.values, 1)
trend_imerg = slope_imerg * x_numeric + int_imerg
trend_gsmap = slope_gsmap * x_numeric + int_gsmap
trend_era5 = slope_era5 * x_numeric + int_era5
# Plot Garis Tren (Dashed Line)
plt.plot(monthly_rain_imerg.index, trend_imerg, color='#004d40', linestyle='--',
plt.plot(monthly_rain_gsmap.index, trend_gsmap, color='#311b92', linestyle='--',
plt.plot(monthly_rain_era5.index, trend_era5, color='#880e4f', linestyle='--', l
plt.title('Perbandingan Tren Curah Hujan Bulanan (2005 – 2025)', fontsize=15, fo
plt.xlabel('Tahun', fontsize=12)
plt.ylabel('Total Curah Hujan Bulanan (mm)', fontsize=12)
plt.grid(True, linestyle=':', alpha=0.6)
plt.legend(loc='upper right', fontsize=10)
plt.tight_layout()
plt.savefig(os.path.join(out_dir, 'monthly_rainfall_trend_2005_2025.png'), dpi=6
plt.show()
```

```
Memuat data jangka panjang 2005 - 2025...
```





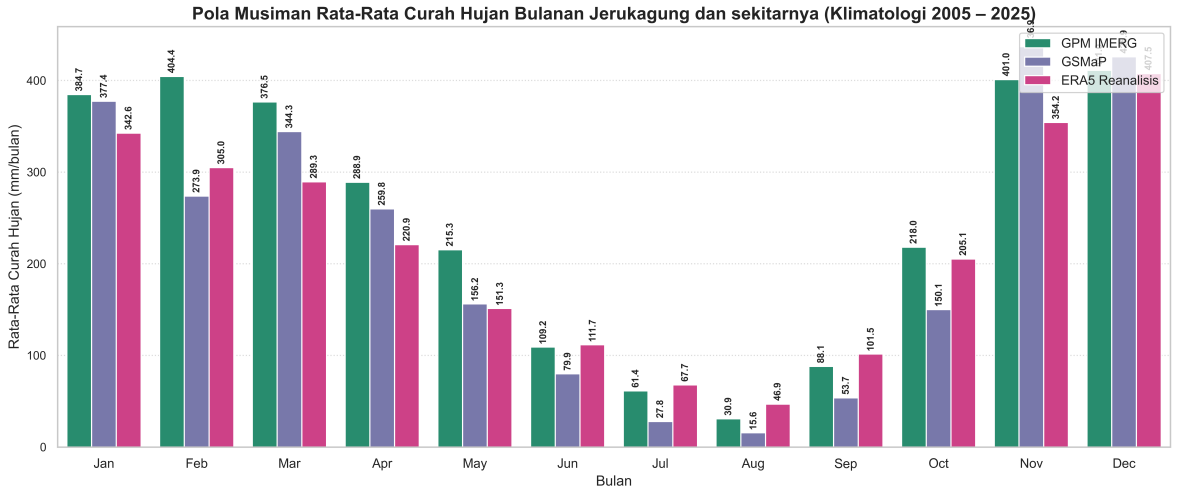
In [24]:
```
## 11. Tren Iklim (2005 – 2025)
```

12. Visualisasi Grafis Temporal Lanjutan

Analisis pola klimatologi bulanan (bar chart), tren perataan harian (moving average), dan variabilitas bulanan (boxplot).

In [25]:
```
# Sel 16.1: Grafik Batang (Bar Chart) Klimatologi Bulanan (Jan - Des)
base_sat_dir = os.path.join(base_dir, 'Data_Satelit') if 'base_dir' in locals()
out_dir = os.path.join(base_dir, 'Hasil_Analisis') if 'base_dir' in locals() els
os.makedirs(out_dir, exist_ok=True)
# Konstruksi DataFrame Bulanan 2005 - 2025
df_m_combined = pd.DataFrame({
    'ERA5': monthly_rain_era5,
    'GSMaP': monthly_rain_gsmap,
    'IMERG': monthly_rain_imerg
})
df_m_combined['Month_Name'] = df_m_combined.index.strftime('%b')
df_m_combined['Month_Num'] = df_m_combined.index.month
# Hitung Rata-Rata Klimatologi Bulanan (2005 - 2025)
df_climatology_list = []
month_names = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'O
for m in range(1, 13):
    sub_m = df_m_combined[df_m_combined['Month_Num'] == m]
    for dataset in ['IMERG', 'GSMaP', 'ERA5']:
        df_climatology_list.append({
            'Bulan_Num': m,
            'Bulan': month_names[m-1],
```

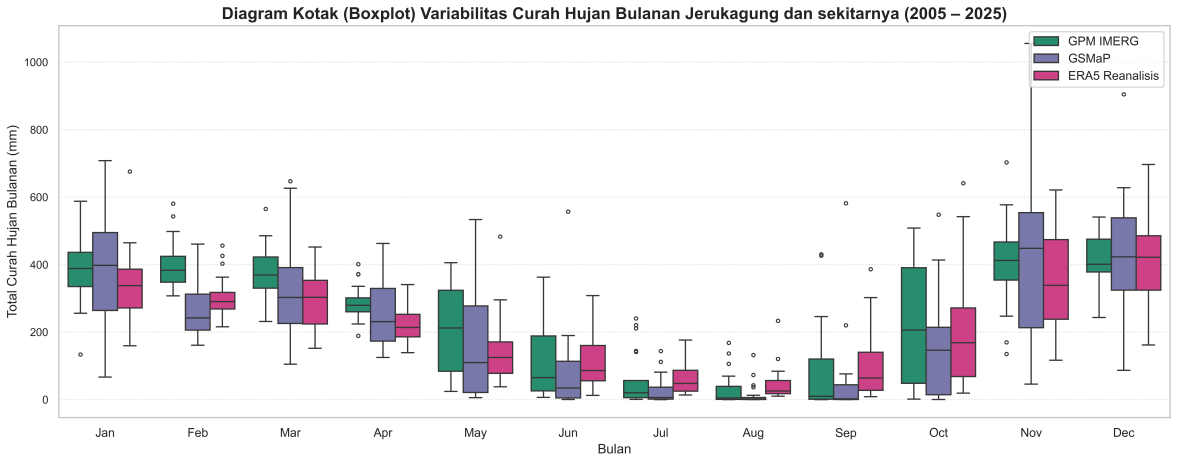
```
            'Dataset': 'GPM IMERG' if dataset=='IMERG' else ('ERA5 Reanalisis' i
            'Curah_Hujan_mm': sub_m[dataset].mean()
        })
df_clim = pd.DataFrame(df_climatology_list)
# Plot Bar Chart Klimatologi Bulanan
plt.figure(figsize=(14, 6))
ax_bar = sns.barplot(
    data=df_clim,
    x='Bulan',
    y='Curah_Hujan_mm',
    hue='Dataset',
    palette=['#1b9e77', '#7570b3', '#e7298a']
)
plt.title('Pola Musiman Rata-Rata Curah Hujan Bulanan Jerukagung dan sekitarnya
plt.xlabel('Bulan', fontsize=12)
plt.ylabel('Rata-Rata Curah Hujan (mm/bulan)', fontsize=12)
plt.xticks(fontsize=11)
plt.yticks(fontsize=10)
plt.grid(True, axis='y', linestyle=':', alpha=0.6)
# Labels di atas setiap bar
for container in ax_bar.containers:
    ax_bar.bar_label(container, fmt='%.1f', fontsize=8, fontweight='bold', paddi
plt.legend(loc='upper right', fontsize=11)
plt.tight_layout()
plt.savefig(os.path.join(out_dir, 'monthly_climatology_barchart_2005_2025.png'),
plt.show()
```



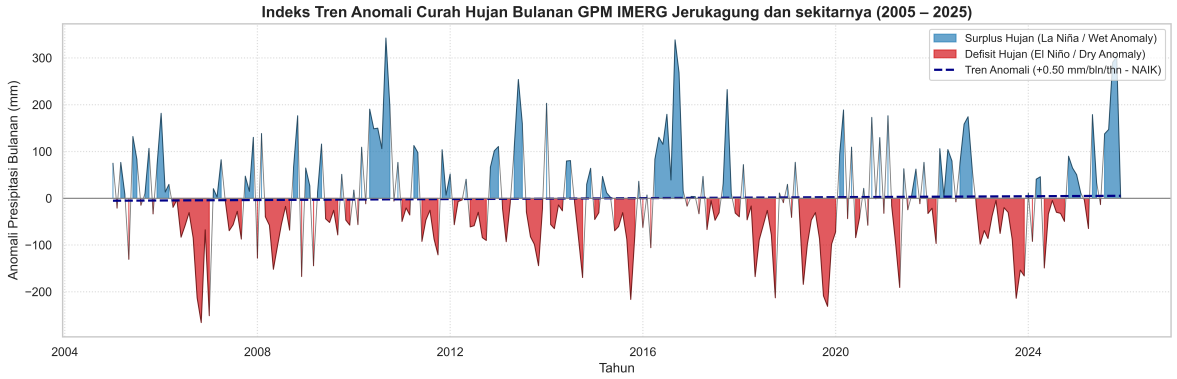
In [26]:
```
## 12. Visualisasi Grafis Temporal
```

In [27]:
```
# Sel 16.3: Diagram Kotak (Boxplot) Temporal Per Bulan (Variabilitas & Outliers)
df_box_data = df_m_combined.melt(
    id_vars=['Month_Num', 'Month_Name'],
    value_vars=['IMERG', 'GSMaP', 'ERA5'],
    var_name='Dataset',
    value_name='Monthly_Rain_mm'
)
df_box_data['Dataset'] = df_box_data['Dataset'].replace({'IMERG': 'GPM IMERG', '
plt.figure(figsize=(15, 6))
ax_box_m = sns.boxplot(
    data=df_box_data,
    x='Month_Name',
    y='Monthly_Rain_mm',
    hue='Dataset',
    palette=['#1b9e77', '#7570b3', '#e7298a'],
```

```
    showfliers=True,
    fliersize=3,
    linewidth=1.2
)
plt.title('Diagram Kotak (Boxplot) Variabilitas Curah Hujan Bulanan Jerukagung d
plt.xlabel('Bulan', fontsize=12)
plt.ylabel('Total Curah Hujan Bulanan (mm)', fontsize=12)
plt.xticks(fontsize=11)
plt.yticks(fontsize=10)
plt.grid(True, axis='y', linestyle=':', alpha=0.6)
plt.legend(loc='upper right', fontsize=11)
plt.tight_layout()
plt.savefig(os.path.join(out_dir, 'monthly_rainfall_variability_boxplot.png'), d
plt.show()
```

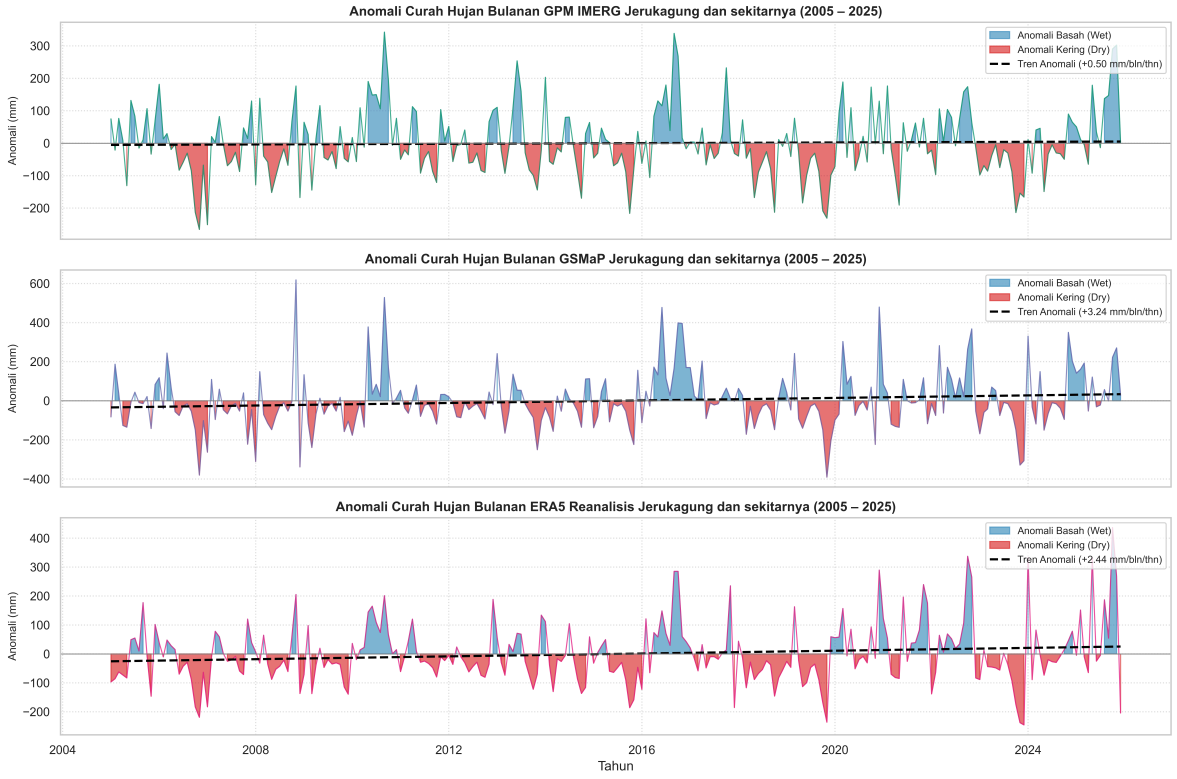


In [28]:
```
# Sel 16.4: Grafik Tren Anomali Curah Hujan Bulanan (Rainfall Anomaly Index 2005
m_imerg = df_imerg_hourly_lt['precipitation'].resample('1MS').sum()

# Hitung Anomali Bulanan (Deviasi terhadap Klimatologi Bulanan)
climatology_m = m_imerg.groupby(m_imerg.index.month).transform('mean')
anomaly_m = m_imerg - climatology_m

plt.figure(figsize=(15, 5))
plt.fill_between(anomaly_m.index, anomaly_m.values, 0, where=(anomaly_m.values >
plt.fill_between(anomaly_m.index, anomaly_m.values, 0, where=(anomaly_m.values <
plt.plot(anomaly_m.index, anomaly_m.values, color='black', linewidth=0.8, alpha=

# Fit garis tren anomali
x_num_anom = (anomaly_m.index - anomaly_m.index[0]).total_seconds() / (86400 * 3
slope_anom, intercept_anom = np.polyfit(x_num_anom, anomaly_m.values, 1)
plt.plot(anomaly_m.index, slope_anom * x_num_anom + intercept_anom, color='darkb

plt.axhline(0, color='gray', linestyle='-', linewidth=1)
plt.title('Indeks Tren Anomali Curah Hujan Bulanan GPM IMERG Jerukagung dan seki
plt.xlabel('Tahun', fontsize=12)
plt.ylabel('Anomali Presipitasi Bulanan (mm)', fontsize=12)
plt.grid(True, linestyle=':', alpha=0.6)
plt.legend(loc='upper right', fontsize=10)
plt.tight_layout()
plt.savefig(os.path.join(out_dir, 'monthly_rainfall_anomaly_trend_2005_2025.png'
plt.show()
```



In [29]:

```
# Sel 16.4: Line Chart Anomali Curah Hujan Bulanan & Tren (2005 - 2025)
# Hitung Anomali Presipitasi Bulanan (Presipitasi Bulanan - Rerata Klimatologi B
clim_mean = df_m_combined.groupby('Month_Num')[['IMERG', 'GSMaP', 'ERA5']].trans
df_anomaly = df_m_combined[['IMERG', 'GSMaP', 'ERA5']] - clim_mean

fig, axes = plt.subplots(3, 1, figsize=(15, 10), sharex=True)
datasets = [('IMERG', 'GPM IMERG', '#1b9e77'), ('GSMaP', 'GSMaP', '#7570b3'), ('

x_dates = df_anomaly.index
x_num = (x_dates - x_dates[0]).total_seconds() / (86400 * 365.25)

for i, (col, title, color) in enumerate(datasets):
    ax = axes[i]
    vals = df_anomaly[col].values
    ax.fill_between(x_dates, vals, 0, where=(vals >= 0), color='#2b83ba', alpha=
    ax.fill_between(x_dates, vals, 0, where=(vals < 0), color='#d7191c', alpha=0
    ax.plot(x_dates, vals, color=color, linewidth=1, alpha=0.8)

    # Fit trendline anomali
    slope_anom, int_anom = np.polyfit(x_num, vals, 1)
    ax.plot(x_dates, slope_anom * x_num + int_anom, color='black', linestyle='--

    ax.axhline(0, color='gray', linestyle='-', linewidth=0.8)
    ax.set_title(f'Anomali Curah Hujan Bulanan {title} Jerukagung dan sekitarnya
    ax.set_ylabel('Anomali (mm)', fontsize=10)
    ax.grid(True, linestyle=':', alpha=0.6)
    ax.legend(loc='upper right', fontsize=9)

plt.xlabel('Tahun', fontsize=12)
plt.tight_layout()
plt.savefig(os.path.join(out_dir, 'monthly_rainfall_anomaly_trend_2005_2025.png'
plt.show()
```


Anomali Curah Hujan Bulanan GPM IMERG Jerukagung dan sekitarnya (2005 – 2025)
Anomali Basah (Wet)
Anomali Kering (Dry)
Tren Anomali (+0.50 mm/bln/thn)
Anomali (mm)
300
200
100
0
−100
−200

Anomali Curah Hujan Bulanan GSMaP Jerukagung dan sekitarnya (2005 – 2025)
Anomali Basah (Wet)
Anomali Kering (Dry)
Tren Anomali (+3.24 mm/bln/thn)
Anomali (mm)
600
400
200
0
−200
−400

Anomali Curah Hujan Bulanan ERA5 Reanalisis Jerukagung dan sekitarnya (2005 – 2025)
Anomali Basah (Wet)
Anomali Kering (Dry)
Tren Anomali (+2.44 mm/bln/thn)
Anomali (mm)
400
300
200
100
0
−100
−200
2004
2008
2012
2016
2020
2024
Tahun